Monetary Policy Transmission to DeFi Yields: Evidence from On-Chain Liquidity and Credit Markets

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Abstract

This study examines whether decentralized finance (DeFi) markets operate as isolated ecosystems or integrate with traditional monetary policy. We investigate how U.S. monetary policy transmits to DeFi yields by comparing two core mechanisms: AMM liquidity provision (Curve 3pool) and decentralized lending (Aave/Compound). Using daily data from 2021–2025, the analysis combines event studies, local projections, and Granger causality tests around FOMC announcements and short-rate movements. Results indicate delayed pass-through in AMM yields, peaking about one week after tightening, consistent with frictions in on-chain liquidity provision. Lending yields display limited event-driven responses and are instead shaped primarily by crypto-native conditions, including leverage demand and market volatility. Monetary tightening is also associated with transitory TVL outflows that revert quickly.

KEYWORDS:

- Decentralized Finance (DeFi)
- Monetary Policy Transmission
- Automated Market Makers (AMMs)
- On-Chain Lending Markets
- Total Value Locked (TVL)

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Contents

1 Introduction **3**

2 Data collection and description of DeFi yields **7**
2.1 Data Collection & Variables 7
2.1.1 Data Sources and Definitions 7
2.2 Yield from Automated Market Makers: The Case of Curve Finance 8
2.2.1 Curve Finance Mechanics 9
2.3 Yield from Credit Markets: Aave and Compound 10
2.3.1 Aave and Compound Protocols 10
2.3.2 Interest Rate Determination 11

3 Quantitative analysis **13**
3.1 LP 3pool Curve 13
3.1.1 Theoretical Framework: The Liquidity Premium in Traditional and Decentralized Finance 13
3.1.2 A Three-Act Story of DeFi's Integration 15
3.1.3 Event Study Methodology: A Dual Approach 16
3.1.4 Determinants of the Curve 3pool Yield 21
3.2 Stablecoin 23
3.2.1 Event Study: The Muted Reaction of DeFi Lending Rates 23
3.2.2 Determinants of the DeFi Lending Yield 25
3.3 Transitory Liquidity Pressures and Genuine Capital Flight: An Event Study on Aggregate TVL 26
3.3.1 Motivation and Methodological Challenges 26
3.3.2 Empirical Findings: A Significant but Transitory Effect 27
3.4 Granger Causality Analysis: Regime-Dependent Transmission 29
3.4.1 Methodology 30
3.4.2 Results Across DeFi Mechanisms 31
3.4.3 Comparative Analysis: Three Distinct Transmission Channels 36
3.4.4 Reconciling Statistical and Economic Significance 38

4 Conclusion **41**

5 Appendix **43**
5.1 Summary statistics fo 3pool 43
5.2 Summary statistics for Aave / Compound 43
5.3 Multicollinearity Test (VIF) 44
5.4 Supplementary Results for Event Study Analyses 45
5.4.1 Dynamic Response of Total Value Locked (TVL) 45
5.4.2 Supplementary Analysis: TVL Price Adjustment 46
5.5 Supplementary Event Study Results: Stablecoin Lending 50
5.5.1 Surprise-Based Analysis 50

Chapter 1

Introduction

The intersection between technological progress and financial innovation has accelerated a structural transformation in global markets: the emergence of blockchain-based financial ecosystems. This development transcends the introduction of a new asset class; it represents a systemic evolution in the design and operation of financial systems. By merging cryptography, distributed computing, and programmable contracts, blockchain technology establishes a new institutional framework, capable of reducing information asymmetries, enhancing transparency, and enabling decentralized governance at a scale previously unattainable. Within this broader transformation, Decentralized Finance (DeFi) has emerged as one of the most advanced and impactful applications of blockchain technology. As emphasized by Schär (2021b), DeFi can be interpreted as an alternative financial infrastructure that recreates traditional financial services through open, interoperable, and transparent protocols, while also introducing new composable building blocks such as decentralized exchanges, on-chain debt markets, and blockchain-based derivatives. Built primarily on public blockchains like Ethereum, DeFi eliminates indeed the need for centralized intermediaries by allowing code to enforce financial agreements autonomously. This shift from institution-based to protocol-based finance, marks a fundamental reconfiguration of trust: value is no longer secured by reputation or regulation alone, but by the transparency and immutability of code itself. As such, DeFi represents not only a technological innovation, but a conceptual leap toward a more open, programmable, and globally accessible financial architecture.

As Rahman et al. (2022) argue, the key question one ought to ask before crafting or investigating a new token-based architecture is Why Decentralize? A principled answer begins by recognizing that many vulnerabilities in traditional finance stem not from technical limitations, but from institutional design, including information asymmetries, concentrated control, and agent problems inherent in centralized intermediation. Blockchain offers a technological countermeasure to these social problems: by distributing validation and anchoring transactions to evident ledgers, it reduces single points of failure and replaces ex post trust in institutions with ex ante assurances encoded in open, verifiable rules. On this foundation, businesses built on smart contracts can narrow the domain of human discretion, automating core processes in ways that selectively hard-code secure and ethical business practices thereby algorithmically enforcing compliance constraints that would otherwise rely on manual oversight. Within this architecture, stablecoins play a pivotal role as the operational bridge between decentralized protocols and the broader economy: by maintaining price parity with reference currencies, they furnish a credible unit of account and medium of exchange for on-chain markets, enable efficient settlement and liquidity provision, and extend programmable finance to users who require monetary stability. In parallel, the exploration of central bank digital currencies (CBDCs) signals a public-sector response to the same technological frontier, retaining sovereign issuance and oversight while contemplating programmable settlement, enhanced payment efficiency, and improved monetary transmission. The resulting landscape is not a binary contest between code and institutions, but a negotiation over where credibly neutral automation should supplant judgment, and where

public mandates and accountability must remain essential. Decentralization is not pursued for its own sake; it is justified where it measurably lowers agency costs, curbs corruption risk, and delivers auditable, rule-based finance while coexisting, where appropriate, with institutional frameworks such as stablecoin governance and CBDC design that secure systemic reliability at scale.

From an investor point of view, the proliferation of cryptocurrencies and DeFi assets has introduced new financial alternatives for investors seeking to optimize their portfolios. Leveraging the principles of the modern portfolio theory (MPT), efficient diversification can be achieved if the correlations between the assets in a portfolio are sufficiently negative (Markowitz (1952)). Thus, it is essential to measure the interrelationships between diverse financial instruments in order to better inform optimal investment decision-making. By appreciating the interconnectedness between various financial instruments, investors can better assess the impact of market shocks on their portfolios and employ effective risk-management strategies (Brøgger (2022)). This observation motivates the distinction between a diversifier and a safe haven: following Baur and Lucey (2010), a safe-haven asset is one that is uncorrelated or negatively correlated with a reference asset during periods of market stress. The empirical evidence on whether crypto-assets qualify as safe havens is mixed. Bitcoin, for instance, often provides limited safe-haven properties and may even increase downside risk during crisis episodes Bouri et al. (2017), an ambiguity reflected in recent systematic reviews. In this thesis, we extend this "safe haven vs. market integration" question from crypto prices to yield-generating DeFi positions, asking whether DeFi yields remain insulated from monetary tightening or instead co-move with the interest-rate cycle.

Like all technologies, decentralized finance protocols involve inherent trade-offs. Blockchain represents a foundational technology aimed at restructuring incentive mechanisms around digital autonomy and financial privacy, rather than serving as a comprehensive solution for the global financial system. As such, investors, developers, and policymakers should understand both the potential and limitations of this new architecture.

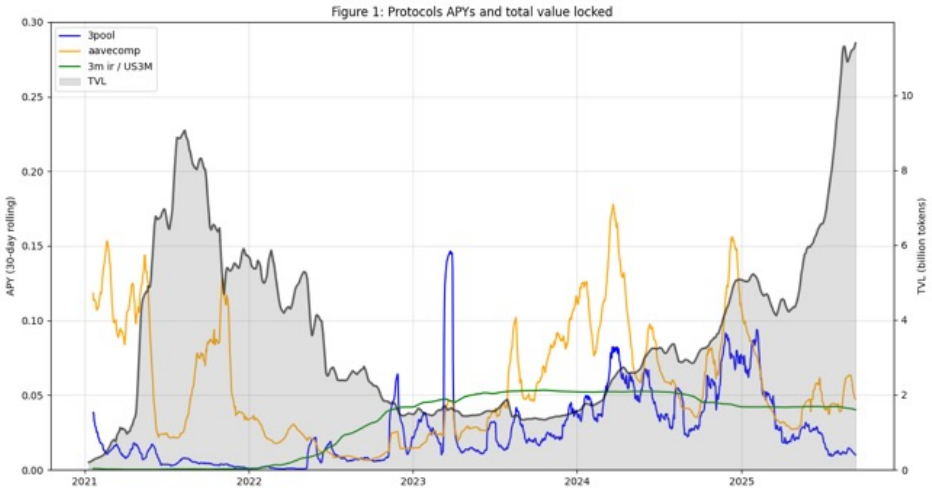
Most DeFi applications remain highly experimental and should be regarded as such. Smart contracts, however carefully designed, rely on assumptions about user behavior that may not hold in practice. When speculative activity dominates over organic adoption, historical data and theoretical models offer limited guidance on practical utility or market stability. Furthermore, algorithmic governance introduces novel challenges: protocols become rigid once deployed, and automated operations lack many of the human checks and balances that stabilize traditional finance. These considerations are not arguments against DeFi innovation, but rather reminders that understanding the underlying mechanics and limitations of decentralization is essential for navigating this evolving space responsibly.

This thesis contributes to a growing literature exploring the interaction between decentralized finance and macro-financial conditions, particularly the extent to which monetary policy transmits to on-chain markets. Our empirical framework builds closely on Barbon et al. (2023), which study the transmission of U.S. monetary policy shocks to stablecoin lending rates. We extend their analysis by examining an additional and structurally distinct DeFi yield generator, liquidity provision in Curve's 3pool, where returns arise from Automated Market makers based trading fees rather than from over-collateralised lending. Specifically, we investigate how these different DeFi yield-generation mechanisms respond to changes in traditional monetary policy, with the objective of assessing whether DeFi assets behave as an independent liquidity ecosystem or whether they are increasingly integrated with the broader macro-financial cycle.

Using daily data extracted from Dune Analytics, major CEX derivatives markets, and traditional financial datasets, we combine cross-sectional regressions, a dynamic event-study methodology around FOMC announcements, and structural pass-through analysis to measure the sensitivity of each DeFi yield to both unanticipated policy shocks and realized tightening of financial conditions. The overarching goal is to determine whether the integration of DeFi with the risk-

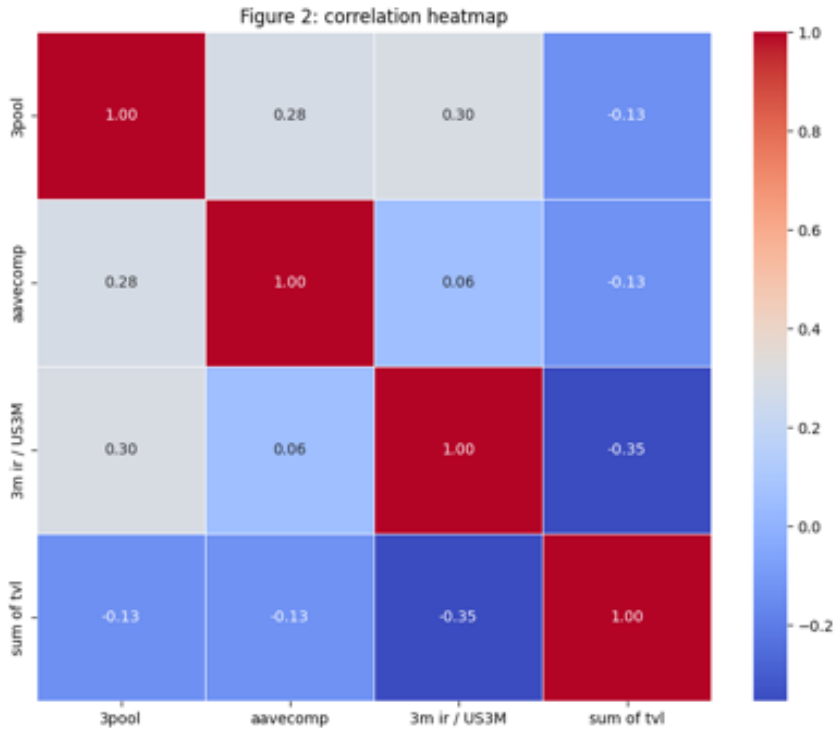
free rate depends on its underlying economic mechanism - market-making (Curve) or leveraged borrowing (Aave/Compound) - and to evaluate whether these increasingly sophisticated protocols behave more like autonomous crypto-native markets or as extensions of the global liquidity cycle. As a proxy for the risk-free rate, this study utilizes the U.S. 3-month Treasury bill interest rate. This choice is consistent with a vast body of academic literature in finance and economics, as it represents the yield on a zero-credit-risk asset and reflects short-term monetary policy expectations. An alternative modern benchmark is the Secured Overnight Financing Rate (SOFR), which reflects the cost of overnight borrowing collateralized by Treasury securities. While the SOFR has become the primary reference rate for U.S. financial markets, we proceed with the 3-month T-bill rate for two main reasons: first, its longer data availability provides a more consistent historical series; second, its 3-month tenor aligns well with the short-to-medium term investment horizons often considered in both traditional and decentralized finance. As a robustness check, one could replicate the analysis using the SOFR, but we leave this for future research.

Figure 1.1 provides a preliminary overview of DeFi yield dynamics relative to both U.S. short-term interest rates and on-chain liquidity conditions. Plotting Curve and Aave/Compound APYs alongside the aggregate USDC+USDT total value locked highlights how DeFi yields fluctuate with changes in available liquidity and broader monetary conditions. Including this figure is relevant because it visually motivates the central research question of the thesis, namely whether DeFi yields constitute an autonomous return structure or whether they adjust systematically to shifts in macro-financial factors and endogenous liquidity cycles within the DeFi ecosystem.



Total value locked (TVL) acts as a macro-level liquidity indicator for the DeFi ecosystem, reflecting the aggregate pool of stablecoin capital available within protocols. Consistency with the view of DeFi as a liquidity and settlement infrastructure in which yields emerge from the supply and demand for on-chain liquidity (Schär (2021a)), higher TVL is expected to be associated with lower protocol yields, whereas liquidity contractions are expected to coincide with higher yields. Figure 1.1 provides an initial visual context for this mechanism by jointly displaying protocol APYs and aggregate TVL over time. In Figure 1.2, we complement this qualitative overview with a correlation heatmap to examine whether the data exhibit patterns consistent with this theoretical liquidity–yield trade-off.

Figure 1.2 provides descriptive evidence consistent with two stylized facts. First, aggregate stablecoin TVL is negatively correlated with the 3-month U.S. Treasury rate (-0.35), suggesting that tighter external monetary conditions are associated with a contraction of on-chain dollar



liquidity. Second, protocol yields display only weak-to-moderate unconditional correlations with both the policy benchmark and TVL (e.g., 3pool-US3M: 0.30; Aave/Comp-US3M: 0.06; yields-TVL: -0.13), indicating heterogeneous and potentially regime dependent drivers across AMM-based and lending-based return mechanisms. Taken together, these patterns motivate the thesis' central analysis of whether DeFi yields constitute an autonomous return structure or adjust systematically to macrofinancial conditions through liquidity reallocation channels.

Chapter 2

Data collection and description of DeFi yields

2.1 Data Collection & Variables

This chapter details the data sources and construction methods for all variables used in our empirical analysis. Our sample period spans from January 1, 2021, to September 11, 2025, and all data is collected at a daily frequency unless otherwise specified. The selection of variables is guided by economic theory and the existing literature on monetary policy transmission and DeFi market dynamics.

2.1.1 Data Sources and Definitions

The primary data on DeFi protocols were collected from on-chain sources via custom SQL queries executed on the Dune Analytics platform, which provides comprehensive access to decoded blockchain data. In terms of return measures, we rely on Annual Percentage Yield (APY). APY is the standard measure of returns in DeFi protocols and reflects the annualized, compounded return earned on an asset. APY captures all protocol-level sources of yield—including lending interest, trading fees, and staking rewards—and therefore provides a comparable metric across heterogeneous mechanisms. To ensure robustness, each APY series was winsorized at the 1st and 99th percentiles prior to index construction, reducing the influence of extreme outliers and protocol-specific data artifacts.

To construct our dependent variables, we focus on two distinct types of DeFi yields. First, we measure a DeFi money-market yield using the Curve 3pool. This series is sourced from on-chain data and captures the return to liquidity provision in a stablecoin-to-stablecoin automated market maker. The daily yield is computed as the trading fees generated within the pool relative to its Total Value Locked (TVL), thereby reflecting the compensation earned by liquidity providers for facilitating swaps. Second, we proxy DeFi credit-market conditions by constructing TVL-weighted lending-rate indices for stablecoin supply yields. Specifically, we build two indices, one for USDC and one for USDT, along with an aggregate index. For each stablecoin, we collect daily supply-side APYs and the corresponding total supplied value in USD for the leading lending protocols on Ethereum, including Aave (V2 and V3) and Compound (V2 and V3). The daily index value is calculated as the weighted average of pool-level APYs, where each pool's weight is its share of TVL across all included protocols for that stablecoin. This aggregation reduces protocol-specific noise and delivers a representative market-wide measure of DeFi lending rates.

To account for crypto-internal dynamics (Crypto-Native control variables) that may jointly drive yields and respond to monetary conditions, we include a set of crypto-native control variables. Speculative leverage demand is proxied by the daily average funding rate of the ETH–USDT perpetual futures contract, sourced from OKX via the CCXT library; higher fund-

ing rates reflect stronger demand for leveraged long exposure to ETH, the primary collateral asset in DeFi. Market frictions are captured by the average daily Ethereum gas price (in Gwei), obtained from a public repository of aggregated on-chain data, which proxies time-varying transaction costs and potential limits to arbitrage between DeFi and traditional markets. Broader crypto-market sentiment and risk conditions are controlled for using Bitcoin's daily logarithmic return and its 30-day rolling historical volatility, sourced from public market-data providers.

Federal Funds Rate (EFFR), sourced from the Federal Reserve Economic Data (FRED) database (ticker: DFF). Because the policy rate changes discretely, we construct a daily series by forward-filling values between official adjustments. As a robustness check, we also employ the 3-month Treasury bill rate (ticker: DGS3MO). To isolate the unanticipated "news" component of FOMC announcements, we use the high-frequency monetary policy shock series of Acosta et al. (2024), constructed from SOFR futures price changes in a 30-minute window around FOMC statements. We employ both the GSS target and GSS path factors.

Finally, we include two global risk and market controls. Global risk aversion is proxied by the CBOE Volatility Index (VIX), sourced from FRED (ticker: VIXCLS), while broad equity-market conditions are controlled for using the daily return of the S&P 500 index from public market-data providers.

2.2 Yield from Automated Market Makers: The Case of Curve Finance

One of the most popular sources of yield in DeFi is providing liquidity in decentralized exchanges. Decentralized exchanges are protocols that allow peer-to-peer trading of assets without centralized intermediaries. DEXs remove the trusted intermediary: users trade directly from their wallets, and smart contracts perform swap settlements, eliminating custody and counterparty risks. The breakthrough in DEX design came with Automated Market Makers (AMMs), popularized by Uniswap. An AMM is essentially a constant function market maker a smart contract that holds reserves of two (or more) tokens and quotes prices algorithmically according to a formula. The most common formula is the constant product rule: for a two-asset pool, $x \cdot y = k$ (a constant). This formula produces a convex price curve where the price of token X in terms of Y rises as X's reserve gets depleted, and vice versa. In practical terms, an AMM always offers a price for a trade, but the more you try to buy of one token relative to the pool size, the more expensive it becomes (slippage). A small trade causes a minor price shift, a large trade causes a bigger shift. This design means the liquidity pool can never be completely drained: as one side approaches zero, its price asymptotically approaches infinity. Thus, AMMs provide continuous liquidity: traders can execute at least a partial trade at any time, as opposed to order books which can have no orders at a given price.

Liquidity on AMM-based DEXs is supplied by liquidity providers (LPs), users who deposit tokens into a liquidity pool in exchange for pool tokens that represent a pro-rata claim on the pool's reserves and fee income. LPs earn trading fees as compensation for making capital continuously available to traders. However, LPs in AMMs face a unique risk of impermanent loss. Because the AMM's formula forces the portfolio to continuously rebalance (selling the outperforming asset to buy more of the underperforming asset to maintain $x \cdot y = k$), an LP's asset holdings can be worse off compared to simply holding the assets outside the pool. The loss is "impermanent" as long as the LP does not withdraw: if prices revert to the original ratio, the loss disappears. But if the LP withdraws liquidity when prices have substantially diverged, the loss is realized (permanent). The fee earnings can sometimes offset IL (Impermanent Loss), but in volatile markets with large price swings, IL can outweigh fee revenue, meaning LPs end up with a lower value portfolio. This is a critical aspect of AMM economics: LPs effectively assume price risk (like short volatility exposure) in exchange for earning transaction fees.

2.2.1 Curve Finance Mechanics

Curve Finance is a decentralized exchange specialized in stablecoin trading and similarly pegged assets. It was launched in 2020 with the goal of optimizing swaps between assets that should maintain the same value and it has become one of the most liquidity-rich protocols in DeFi.

The focus of our analysis will be the Curve 3Pool, a liquidity pool containing the three major USD stablecoins: DAI, USDC, and USDT. The 3Pool serves as a base pool providing deep, capital-efficient liquidity for swapping between these top stablecoins. This stablecoin specialization made Curve 3Pool critical infrastructure for DeFi strategies, it became the go-to platform whenever users or protocols needed to exchange or park stable-value assets efficiently.

The major innovation introduced with the Curve 3Pool in Egorov (2019) (the original white paper), is a specifically designed AMM formula, which differs from the standard ($x \cdot y = k$) model, for assets that should trade around the same price:

$$An^n \sum_i x_i + D = ADn^n + \frac{D^{n+1}}{n^n \prod_i x_i} \tag{2.1}$$

where D is the invariant (analogous to k in other AMMs); $n = 3$ in 3Pool, and A is the amplification coefficient.

Instead of a pure constant-product formula, Curve combines a constant-sum component (which would give zero slippage at 1:1) with a constant-product component, which guarantees well-behaved prices when the pool is imbalanced. The key to low slippage is the parameter A. When the pool is roughly balanced, the term multiplied by A dominates, and the curve behaves almost like a constant-sum AMM: the marginal price stays very close to 1:1 for relatively large order sizes, so the change in the output amount per additional unit traded (i.e., slippage) is extremely small. As the pool becomes more imbalanced, say one stablecoin is heavily withdrawn, A falls, the constant-product term becomes more important, and the invariant smoothly transitions toward constant-product behaviour. This prevents the pool from offering "infinite" liquidity at 1:1 and restores stronger price sensitivity that incentivizes arbitrageurs to rebalance the pool. In practice, 3Pool sets a high A so that, around the equilibrium composition of DAI/USDC/USDT, the pricing curve is very flat and thus slippage for stablecoin swaps is much lower than in a standard constant-product AMM, while still remaining stable and well-behaved under larger imbalances.

The APY earned by Curve 3Pool Liquidity Providers is generated through the accumulation and compounding of swap fees and, historically, gauge-based incentive rewards. Every trade executed within the pool pays a fee which is pro-rata distributed to LPs by increasing the pool's underlying stablecoin balances. Because the StableSwap invariant keeps slippage extremely low around the peg, 3Pool attracts high routing volume from aggregators, which mechanically increases fee accrual. LPs are compensated in 3Crv, a tokenized claim representing their share of the pool; its value appreciates over time as the underlying balances grow, creating an implicit compounding yield. When gauge incentives are active, LPs may also receive CRV emissions proportional to their staked liquidity, which can further be boosted through voting-escrow (veCRV) participation. The resulting APY is therefore the combination of realized trading-fee revenue embedded in the rising redemption value of 3Crv, and of any additional token incentives, both of which depend on market volatility, stablecoin demand flows, and the pool's depth and balance.

The 3Pool APY series is mostly low and stable, which is what we expect when the pool is balanced and trading activity is moderate. Every so often, the APY jumps sharply: these smaller spikes usually happen when one stablecoin becomes slightly more demanded than the others, creating temporary imbalances that increase swap fees for liquidity providers. The huge spike in early 2023 is the clearest example of this mechanism. It corresponds to the USDC depeg during the Silicon Valley Bank crisis, when users rushed to exit USDC, the pool became almost entirely filled with it, and trading volumes surged to billions of dollars in a few hours. Under these extreme conditions, Curve's AMM formula becomes highly sensitive to imbalance, so fee

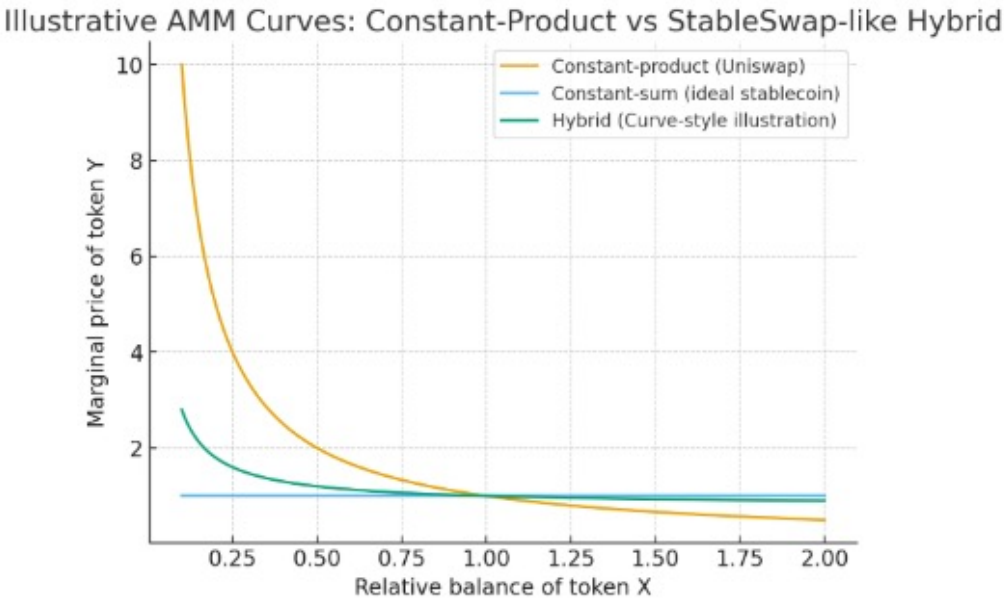


Figure 2.1: Comparison of AMM Curves

generation explodes and the resulting APY looks disproportionately large. The other spikes in 2024–2025 follow the same logic: they reflect short-lived episodes of stress or unusual stablecoin flows that temporarily push the pool away from equilibrium and make yields jump.

2.3 Yield from Credit Markets: Aave and Compound

In DeFi lending protocols, smart contracts replace key intermediation functions traditionally performed by banks through transparent, rule-based execution. The architecture is typically open and permissionless: any user can supply liquidity or take a loan by interacting directly with the code, without identity checks or credit approval. To make this trust-minimized model viable, DeFi loans require overcollateralization. In other words, borrowers must lock crypto assets of higher value than the loan as collateral into the smart contract, which holds it escrowed until the loan is repaid. This design mitigates counterparty risk in an anonymous setting: if a borrower fails to repay, the protocol can seize and sell the collateral to cover the debt. In a pooled design, lenders deposit assets into a common smart contract pool and receive an interest baring receipt token (e.g. cTokens or aTokens) representing their claim, and borrowers can draw from this liquidity provided they post sufficient collateral. Because all lenders and borrowers interact with the same pool, interest rates are determined algorithmically by supply and demand for each asset. When funds are abundant and utilization of the pool is low, the borrowing interest rate is low; if most of the pool's assets are lent out (high utilization), interest rates climb, incenting more liquidity to enter or some borrowers to repay. Through this mechanism, interest rates adjust in real time to balance the market, rather than being fixed by a central authority. The following section describe in detail how these mechanisms are implemented in Aave and Compound.

2.3.1 Aave and Compound Protocols

Aave and Compound are decentralized money market protocols that facilitate lending and borrowing through pooled liquidity, rather than direct peer-to-peer loans. In these systems, lenders deposit assets into a common reserve pool, and borrowers can draw loans from this pool, as long as they post sufficient collateral. This lending pool design means individual deposits be-

come part of a fungible liquidity pool instead of being tied to any single borrower. As a result, withdrawals are on-demand unless the pool's liquidity is fully utilized; depositors can reclaim their funds at any time without waiting for a specific loan to mature. Each asset supported by Aave or Compound has its own market (pool) with a transparent ledger of supply, borrow, and interest rate information. Neither Aave nor Compound relies on a central counterparty or order book to match lenders and borrowers. Instead, all interactions are handled by smart contracts on-chain, making these protocols non-custodial and automated. Users remain in control of their funds via their wallets, and loan enforcement is achieved through protocol rules (e.g., collateral requirements and liquidation mechanisms) rather than legal contracts.

In Aave and Compound, depositors earn interest income paid by borrowers. When someone borrows from the pool, they accrue interest on their loan balance at the current variable rate; this interest is distributed to the suppliers of that asset (proportional to their share of the pool) as yield. The protocols charge borrowers a slightly higher rate than what is paid to suppliers, reserving a fraction of interest (the reserve factor) to accumulate insurance reserves. Still, effectively, deposit APYs are generated from borrower interest payments, making these platforms analogous to interest-bearing savings accounts or bond markets in DeFi.

Both protocols implement tokenized representations of deposits that embody the lender's principal plus accrued interest. In Compound, when a user supplies assets, the protocol mints cTokens (e.g., cDAI, cETH) to the supplier's address. cTokens are standard ERC-20 tokens that appreciate in value relative to the underlying asset: each cToken has an exchange rate (initially, say, 1 cDAI = 0.02 DAI) that increases over time as borrowers pay interest into the pool. For example, if the cDAI exchange rate grows from 0.02 to 0.021 DAI, a user holding 100 cDAI can now redeem it for 2.1 DAI (instead of 2.0 initially).

The exchange rate formula is:

$$\text{Exchange Rate}_{cToken} = \frac{\text{Cash} + \text{Total Borrows} - \text{Reserves}}{\text{cToken Supply}} \tag{2.2}$$

Simply holding cTokens will automatically accumulate yield, and these tokens can also be used as collateral themselves for borrowing other assets. Aave takes a slightly different but conceptually analogous approach with aTokens (e.g., aUSDC, aETH). When a user supplies assets to Aave, they receive an equal amount of aTokens (1:1 to the deposited asset). These aTokens are interest-bearing in that the quantity of aTokens in the user's wallet increases over time, reflecting earned interest (rather than the token-to-underlying exchange rate changing). In effect, Aave aTokens are like a rebasing token or an accounting entry: if the market's interest rate generates, say, 5% annual yield, a user who deposited 100 USDC will see their aUSDC balance grow to ≈ 105 aUSDC after one year.

In both cases, the protocols ensure that the value of deposits grows in tandem with the interest paid by borrowers. The yield for liquidity providers is thus an automatic and continuous process: interest is accrued by borrowers and periodically allocated to lenders.

2.3.2 Interest Rate Determination

Both Aave and Compound determine interest rates algorithmically through interest rate models that respond to the utilization ratio U. The utilization ratio is the core input to define the interest rate. Utilization is defined as $U = \frac{\text{Total Borrows}}{\text{Total Liquidity}}$, i.e., the percentage of available liquidity that is currently lent out. When utilization is low, the borrow interest rate is kept low to encourage borrowing; when utilization is high, the rate increases, making borrowing more expensive and lending more attractive, so as to draw the system back toward equilibrium.

Practically, Aave and Compound use a piecewise-linear ("kinked") interest rate model in each market. They define a target optimal utilization U^* (or "kink" point). For utilization below U^*, the borrow rate increases gradually (linear slope); once utilization exceeds U^*, the

borrow rate grows much more steeply to penalize reaching full utilization. Mathematically, one can express a typical borrow rate function as:

$$i_b(U) = \begin{cases} i_0 + U \cdot \text{slope}_1, & U \leq U^*; \\ i_0 + \text{slope}_1 + \frac{U-U^*}{1-U^*} \cdot \text{slope}_2, & U > U^*, \end{cases} \tag{2.3}$$

Where slope_1 is the interest rate coefficient below the kink and slope_2 is a larger coefficient for the region above the kink. Aave's and Compound's specific parameters differ by asset, but both employ this model to balance liquidity. The supply interest rate for depositors is derived from the borrow rate and utilization: suppliers earn the weighted rate that borrowers pay.

In Compound, the exact relationship is:

$$i_{supply} = i_{borrow} \cdot U \cdot (1 - \text{Reserve Factor}) \tag{2.4}$$

meaning if the pool is half utilized, lenders earn roughly half the borrow Annual Percentage Rate (after a reserve cut). This implicitly rewards lenders only when their capital is actually being used, and ensures the protocol accrues a reserve buffer.

Key difference between Aave and Compound is that Aave's interest rate model offers borrowers a choice between variable (floating) rates as above, or stable rates for certain assets. A variable rate loan in Aave will fluctuate block-by-block as the pool's rate changes, much like Compound's loans. A stable rate loan in Aave locks in a semi-fixed interest rate for the borrower, providing more predictability.

Chapter 3

Quantitative analysis

3.1 LP 3pool Curve

3.1.1 Theoretical Framework: The Liquidity Premium in Traditional and Decentralized Finance

To understand the yields earned by liquidity providers (LPs) in the Curve 3pool, it is helpful to start with a familiar concept from traditional finance: the liquidity premium. As Nagel (2016) describes, this premium captures the yield discount that investors willingly accept when they demand highly liquid assets, like short-term Treasuries, which offer unique benefits such as easy collateral use and quick settlements during market turmoil. In contrast, the returns from Curve's 3pool represent something of an inversion. A supply-side premium that rewards investors for providing liquidity in a decentralized setting, primarily through trading fees (and, when applicable, incentives), while bearing AMM-specific risks. Both phenomena, however, share a common thread: the opportunity cost of capital, driven by the short risk-free rate (proxied by T-bill yields). In this section, we will explore these distinctions, formalize the connections, and derive predictions for how DeFi yields might respond to shifts in traditional monetary policy.

The Traditional Liquidity Premium: Why Investors Pay for Liquidity

At its core, Nagel's liquidity premium $\lambda_t^{\text{Demand}}$ explains why safe, liquid assets like Treasury bills often yield less than comparable alternatives. Imagine an investor deciding between a Treasury bill yielding R_t^{TB} and a slightly riskier but higher-yielding term deposit at R_t^{Illiquid}. The choice reduces to the non-pecuniary benefits of liquidity: Treasuries aren't just investments; they're tools for efficient transactions and buffers against stress. The investor is indifferent when the higher yield on the illiquid option exactly offsets these advantages:

$$R_t^{\text{Illiquid}} = R_t^{\text{TB}} + \lambda_t^{\text{Demand}}, \tag{3.1}$$

where $\lambda_t^{\text{Demand}} > 0$ quantifies that "yield sacrifice." Building on this, Krishnamurthy and Vissing-Jorgensen (2012) how the premium fluctuates with the scarcity of safe assets and, crucially, with interest rate levels. When central banks hike rates, the foregone earnings from parking money in liquid holdings grow, making the premium larger. Empirically, a 100 basis point rise in the federal funds rate widens the T-bill/repo spread by 10–15 basis points Nagel (2016). This gives us our first key insight: higher R_t^f amplifies the demand for liquidity, pushing $\lambda_t^{\text{Demand}}$ upward:

$$\frac{\partial \lambda_t^{\text{Demand}}}{\partial R_t^f} > 0. \tag{3.2}$$

It's a demand-driven story, played out in institutional markets by cash managers and repo desks chasing safety.

Flipping the Script: Supplying Liquidity in the Curve 3pool

The DeFi ecosystem, by contrast, operates through an inverted mechanism whereby participants provide rather than consume liquidity, facilitating low-slippage swaps. These Liquidity Providers hold inventories and absorb trades, earning compensation through fees rather than spreads. For an LP committing capital K_t to the pool, the expected return R_t^{LP} breaks down into fees from trading volume V_t, a small drag from impermanent loss $\phi(\sigma_t^2)$, and protocol incentives Ψ_t like token rewards:

$$R_t^{\text{LP}} = \frac{f \cdot V_t}{K_t} - \phi(\sigma_t^2) + \Psi_t, \tag{3.3}$$

where $f = 0.04\%$ is the swap fee. Impermanent loss, which can erode returns in volatile pools, is negligible here: with assets pegged at 1:1, $\phi \approx 0$, and even during the 2023 USDC de-peg (to \$0.88), it stayed below 1.00% thanks to the invariant's flat curve Adams et al. (2021). In essence, Curve LPs face low directional risk, trading it for blockchain-specific hazards like smart contract vulnerabilities or gas fees.

Uniting the Worlds: The Opportunity Cost Channel

To keep LPs engaged, DeFi must compensate for the risks and frictions involved, leading to a no-arbitrage condition:

$$\mathbb{E}[R_t^{\text{LP}}] \geq R_t^f + \lambda_{\text{SC}} + \lambda_{\text{gas}}, \tag{3.4}$$

where λ_{SC} covers smart contract risks and λ_{gas} gas costs. This defines the Liquidity Service Premium as the excess yield over the risk-free rate:

$$\lambda_t^{\text{Supply}} = R_t^{\text{LP}} - R_t^f. \tag{3.5}$$

As R_t^f rises, the relative attractiveness of risk-free assets induces liquidity outflows from DeFi, mechanically driving up on-chain yields. This response mirrors the pricing dynamics of traditional markets:

$$\frac{\partial \lambda_t^{\text{Supply}}}{\partial R_t^f} > 0. \tag{3.6}$$

Yet the symmetry is skin-deep. Traditional premia adjust near-instantly via institutional arbitrage, while DeFi's retail-heavy ecosystem introduces delays: on-chain settlements take seconds to minutes, gas optimization encourages batching, and information spreads gradually among individual LPs.

While the theoretical framework emphasizes the liquidity service premium $\lambda_t^{\text{Supply}}$ as the key mechanism, our empirical analysis focuses on the total yield R_t^{LP} for both practical and theoretical reasons. First, retail-oriented DeFi investors do not explicitly decompose returns into risk-free components and premia; rather, they evaluate total returns when comparing opportunities across traditional and decentralized markets. This behavioral reality suggests that R_t^{LP}, as the observable "all-in" yield, is the relevant variable driving capital allocation decisions. Second, analyzing R_t^{LP} captures the full transmission mechanism from monetary policy to DeFi: changes in R_t^f affect DeFi yields both through the baseline opportunity cost and through adjustments in the excess premium $\lambda_t^{\text{Supply}}$. Thus, the correlation between R_t^{LP} and R_t^f reflects the total integration effect, encompassing both channels. Finally, stablecoin pool yields are naturally advertised and tracked as headline APYs rather than spread-based measures, making R_t^{LP} the market-relevant metric for empirical investigation.

3.1.2 A Three-Act Story of DeFi's Integration

Having established the theoretical distinction between traditional and DeFi liquidity premia, we now examine how this relationship manifests in total LP yields across three monetary policy regimes. Our empirical focus on $R_t^{LP}otal$ captures the complete transmission mechanism from traditional yields to DeFi returns.

Formal Testing of Structural Breaks. We test for structural breaks using Chow (1960) tests at two critical junctures: February 2022 (ZIRP to hiking cycle) and August 2023 (hiking cycle to plateau). The test compares residual sum of squares (RSS) from pooled versus separate regressions:

$$F = \frac{(\text{RSS}_{\text{full}} - \text{RSS}_{\text{split}})/k}{\text{RSS}_{\text{split}}/(n-2k)} \tag{3.7}$$

Break 1 (February 2022): $F = 2.81$ ($p = 0.061$) shows marginal significance at 10%. This gradual break is economically intuitive: the Fed implemented 11 rate increases over 16 months rather than a discrete shift, and crypto-specific shocks (UST/Luna May 2022, FTX November 2022) introduced substantial noise, delaying stable DeFi-TradFi correlation.

Break 2 (August 2023): $F = 6.96$ ($p = 0.001$) decisively rejects parameter stability at 1%. The Fed's shift to a prolonged pause allowed convergence on stable opportunity cost expectations. With the macro regime stabilized, the asynchronous integration became structurally embedded.

These results validate our framework: the 2022 transition represents gradual awakening of the opportunity cost channel, while 2023 marks rystallization of asynchronous equilibrium.

Table 3.1: Performance Metrics and Structural Break Tests Across Regimes

Metric	**Zero Interest (Sep'20–Jan'22)**	**Hike Cycle (Mar'22–Jul'23)**	**Plateau (Aug'23–Sep'25)**
Average Annual Return (%)			
3pool	0.97	1.81	4.03
T-Bill ETF	-0.07	2.53	4.81
Annual Volatility (%)			
3pool	0.07	0.33	0.20
T-Bill ETF	0.14	0.30	0.29
Correlation			
3pool vs T-Bill ETF	-0.02	0.08	0.15
Structural Break Tests (Chow, 1960)			
ZIRP → Hike (Feb 2022)	$F = 2.81$ ($p = 0.061$)*		
Hike → Plateau (Aug 2023)			$F = 6.96$ ($p = 0.001$)***

Notes: Returns are annualized arithmetic means of daily percentage changes. Volatility is annualized standard deviation. Chow F-statistics test parameter stability at regime transitions. * $p < 0.10$, *** $p < 0.01$. Marginal 2022 significance reflects progressive Fed tightening and crypto shocks; sharp 2023 break reflects macro stabilization.

Act I: The ZIRP Period (Sep 2020–Jan 2022) – Decoupled World. During ZIRP, the Federal Reserve held rates at 0–0.25% and Treasury yields below 15bp. Table 3.1 shows the 3pool generated 0.97% annually with 0.07% volatility, while the T-Bill ETF returned -0.07% with 0.14% volatility. Correlation of -0.02 confirms complete statistical decoupling.

With negligible opportunity cost, Nagel (2016) traditional Liquidity Premium was dormant,bno meaningful alternative yield existed. The T-Bill ETF mechanically tracked the near-

zero policy rate, while the 3pool's Liquidity Service Premium was driven purely by on-chain factors: trading volume, pool imbalances, and crypto-native volatility.

Act II: The Hike Cycle (Feb 2022–Jul 2023) – Forced Integration. The Fed's 11 rate increases (525bp over 16 months) introduced a rising opportunity cost, igniting a market-wide Liquidity Premium. The T-Bill ETF return surged to 2.53%, while the 3pool doubled to 1.81% with volatility converging to 0.33% (vs 0.30% for T-Bills). Correlation turned positive at 0.08, but the marginal Chow test ($p = 0.061$) indicates noisy, incomplete integration, consistent with gradual learning rather than instantaneous arbitrage.

The 3pool's rising yield reflects the now-significant Liquidity Service Premium: with T-Bills offering 5%+, compensation to attract DeFi liquidity providers, who bear smart contract risk, impermanent loss, and gas costs. The 3pool's idiosyncratic Alpha disappeared, replaced by Beta exposure to traditional factors. However, crypto-specific shocks (UST/Luna, FTX) introduced severe noise, delaying stable transmission. The directional shift is clear: the opportunity cost channel became operative, and DeFi yields began exhibiting sensitivity to the risk-free rate, though with notable frictions.

Act III: The Plateau (Aug 2023–Present) – Asynchronous Equilibrium. With rates held at 5.25–5.50% for 26+ months, both assets deliver strong returns (4.81% vs 4.03%) with increased correlation (0.15). The highly significant Chow test ($F = 6.96$, $p = 0.001$) confirms a genuine structural break.

The relationship has matured into asynchronous equilibrium: the causal link from T-Bills to 3pool strengthened (evidenced by the sharp Chow F-statistic increase), but transmission is no longer instantaneous. Lack of contemporaneous correlation in regressions, combined with significant lagged effects in Granger tests (3.4), shows the 3pool absorbs macro signals with a 7-day delay (peak response in local projections, Figure 3.2). This reflects developing but not fully efficient price discovery, likely due to: information diffusion lags, weekly institutional rebalancing, gas cost optimization, and AMM yields' dependence on accumulated volume rather than instant price adjustments.

Conclusion. This provides initial evidence that a change occurs with the monetary cycle turn. The traditional liquidity premium's influence is now structurally embedded ($F = 6.96$, $p = 0.001$), but integration is asynchronous: traditional shocks transmit with multi-day lags. DeFi has not merely integrated but it has adapted, developing its own distinct, slower price discovery process for valuing liquidity in a high-rate world.

3.1.3 Event Study Methodology: A Dual Approach

To provide a comprehensive analysis of monetary policy transmission, we employ a dynamic event study using a dual-methodological design. This framework allows us to test the efficiency and nature of the DeFi market's reaction to Federal Open Market Committee (FOMC) announcements.

Both approaches rely on a multivariate regression framework estimated over an extended event window of $[-2, +20]$ days.

Abnormal Return Calculation. The Abnormal Return for each day in our event window is calculated using a robust market model. For each FOMC event, we first estimate a "normal" return model on a 60-day estimation window preceding the event (specifically, from $T - 62$ to $T - 3$). This window length was chosen as a balance between statistical stability (ensuring a sufficient number of observations to estimate a reliable baseline) and relevance, as it is short enough to reflect recent market conditions without being contaminated by distant regimes.

Methodology 1: Reaction to Unanticipated News (Surprise-Based). The first approach tests the market's informational efficiency by measuring its response to the unexpected component of FOMC announcements. We regress the daily abnormal return of asset i on day $t + d$ on the high-frequency monetary policy shocks from Acosta et al. (2024):

$$AR_{i,t+d} = \alpha_d + \beta_{1,d}\text{TargetSurprise}_t + \beta_{2,d}\text{PathSurprise}_t + \Gamma'\text{Controls}_{t+d} + \varepsilon_{i,t+d} \tag{3.8}$$

The coefficient of interest, $\beta_{1,d}$, captures the causal impact of the unanticipated monetary policy surprise, testing whether DeFi markets incorporate new macroeconomic information efficiently.

Methodology 2: Reaction to Realized Rate Changes (Pass-Through Analysis). The second, complementary model tests whether the market reacts not to abstract policy news but to the realized tightening or loosening of financial conditions. We regress the daily abnormal returns on the cumulative change in the 3-month Treasury bill rate since the announcement:

$$AR_{i,t+d} = \alpha_d + \gamma_d \Delta^{\text{cum}}\text{RiskFree}_{[t,t+d]} + \Gamma'\text{Controls}_{t+d} + \varepsilon_{i,t+d} \tag{3.9}$$

Here, γ_d measures the dynamic pass-through from realized rate changes to DeFi yields, capturing potential frictions or delays in the transmission process.

Control Variables. To mitigate omitted-variable bias and isolate the causal effects of monetary policy shocks, both models include the comprehensive vector of control variables γ defined in Section 2.1. These controls capture the primary crypto-native forces—speculative leverage demand (ETH perpetual funding rate), market frictions (gas fees), and market depth (TVL)—as well as global macro-financial conditions (VIX and S&P 500 returns). Controlling for these factors ensures that the estimated coefficients on monetary policy variables reflect transmission specific to policy shocks rather than to endogenous crypto-market cycles or shifts in global risk sentiment.

Comparing Surprise-Based vs Pass-Through Methodologies

To test whether DeFi markets respond to informational shocks or to realized financial conditions, we implement both surprise-based and pass-through event studies. Table 3.2 presents the results.

The stark contrast between the two methodologies is immediately apparent. Surprise-based estimates, which isolate the market's reaction to genuinely unanticipated FOMC announcements using high-frequency shocks from Acosta et al. (2024),yield no statistically significant coefficients at any horizon. The magnitudes are an order of magnitude smaller (approximately 10^{-3} versus 10^{-2}), and p-values uniformly exceed conventional thresholds.

In contrast, pass-through estimates exhibit clear statistical significance at horizons of 7-10 days, with economically meaningful magnitudes. This divergence provides direct evidence that DeFi markets do not efficiently incorporate monetary policy "news" within the narrow windows used in traditional high-frequency identification.

Table 3.2: Comparison of Event Study Methodologies: Surprise-Based vs Pass-Through

Horizon	**Surprise-Based**		**Pass-Through**		**Ratio**
	β_{target}	p-value	γ	p-value	
$T-2$	-0.0004	0.577	-0.014	0.710	—
$T-1$	-0.0004	0.108	-0.007	0.820	—
$T=0$	0.0000	—	0.013	0.034**	—
$T+1$	-0.0007	0.270	0.014	0.120	20×
$T+2$	-0.0005	0.056	0.014*	0.045	28×
$T+3$	-0.0005	0.139	0.008	0.340	16×
$T+7$	-0.0019	0.437	**0.025*** **	< 0.001	**13×**
$T+10$	-0.0022	0.700	0.021	0.080	10×
$T+15$	-0.0015	0.399	-0.006	0.650	—
$T+20$	-0.0018	0.269	-0.002	0.870	—
Significant horizons:					
Surprise-based: 0/11					
Pass-through: 3/11					

Notes: β_{target} measures response to unanticipated target surprises from Acosta et al. (2024). γ measures response to cumulative realized changes in 3-month Treasury rate. Ratio shows $|\gamma/\beta_{\text{target}}|$. ***, **, * denote significance at 1%, 5%, 10%.

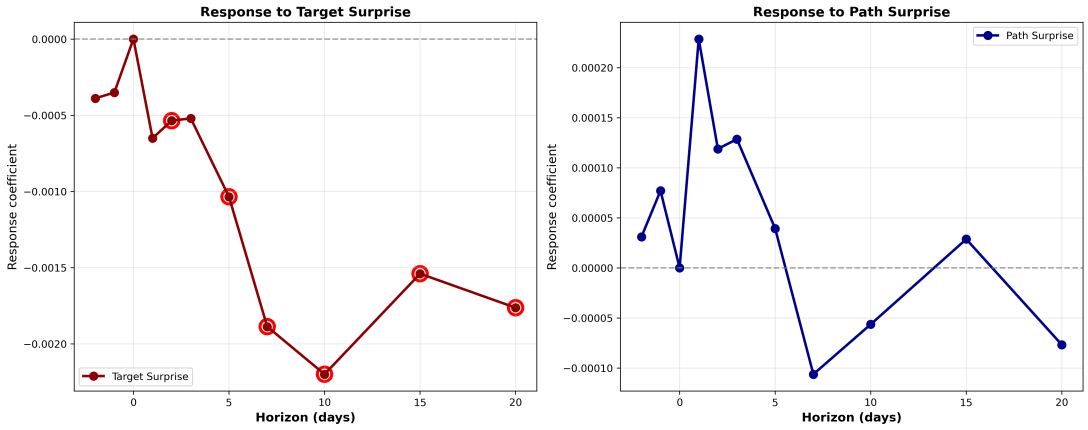


Figure 3.1: Response to Monetary Policy Surprises: Target vs Path Factors. The figure displays impulse responses to unanticipated policy shocks identified using high-frequency changes in SOFR futures Acosta et al. (2024). Red circles indicate horizons where coefficients would be significant at the 5% level. The absence of significant responses contrasts sharply with the pass-through methodology (Figure 3.1), indicating that DeFi markets do not efficiently incorporate high-frequency news.

This finding has three important implications. First, it validates our methodological choice: pass-through analysis is the appropriate framework for studying DeFi-TradFi transmission, as surprise-based methods assume an efficiency that demonstrably does not hold. Second, it confirms the "asynchronous integration" thesis. DeFi is connected to traditional finance, but through slow, observation-driven channels rather than fast, information-driven arbitrage. Third, it reveals the composition of DeFi market participants: the absence of surprise-responses suggests limited participation by sophisticated, high-frequency actors who dominate TradFi derivatives markets.

Third methodology: Local Projections

For each horizon $h \in \{1, 2, 3, 5, 7, 10, 15, 20, 30\}$ days, we estimate:

$$y_{3pool,t+h} - y_{3pool,t} = \alpha^h + \beta^h(r_{f,t+h} - r_{f,t}) + \Gamma^h X_t + \varepsilon_{t+h} \tag{3.10}$$

where the dependent variable is the cumulative change in the 3pool yield from the FOMC announcement date (t) to horizon h, the key regressor is the corresponding cumulative change in the 3-month Treasury rate, and X_t represents our standard set of crypto-native and macroeconomic controls measured at the announcement date. We restrict estimation to FOMC event dates, ensuring that observed rate changes reflect genuine monetary policy actions rather than noise in daily fluctuations. Standard errors are computed using Newey-West HAC estimation with lag length scaled to the horizon h.

This approach offers two methodological advantages over static regressions or single-horizon event studies: first, it captures the full dynamic trajectory of adjustment without imposing a specific lag structure; second, it distinguishes transient responses from permanent shifts, allowing us to reconcile short-run and long-run effects.

Empirical Findings: A Three-Phase Response

Table 3.3 and Figure 3.2 present the estimated response coefficients. The results reveal a striking temporal pattern that reconciles the seemingly contradictory findings from earlier sections.

Table 3.3: Robustness Check: Local Projections With vs Without Controls

	With Controls		**Without Controls**		
Horizon	β	p-val	β	p-val	**Interpretation**
1	-0.014	(0.712)	-0.007	(0.677)	Mediation
2	+0.014	(0.103)	+0.013**	(0.031)	Robust
3	+0.003	(0.389)	+0.003	(0.394)	Robust (Null)
5	+0.008*	(0.074)	+0.001	(0.772)	Suppression
7	**+0.025*** **	**(< 0.001)**	**+0.024*** **	**(0.005)**	**Robust (Core)**
10	+0.021	(0.172)	+0.034**	(0.032)	Mediation
15	-0.006	(0.492)	-0.001	(0.935)	Weak
20	-0.001	(0.978)	+0.001	(0.939)	Weak
30	-0.002	(0.476)	-0.006**	(0.014)	Spurious w/o Controls

Note: The table compares response coefficients (β^h) from the full model (With Controls) against the parsimonious model (Without Controls). Exact p-values are reported in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Phase I: Immediate Immunity ($h = 1 - 3$ days). Contrary to the hypothesis of instantaneous flight-to-quality, we find no statistically significant response in the immediate aftermath of FOMC announcements. The point estimate at $h = 1$ is actually slightly negative (-0.014), but with a p-value of 0.71, this is indistinguishable from zero. This "immunity" to immediate shocks reflects the mechanical nature of AMM yield generation: the 3pool's returns depend on accumulated trading fees, which do not adjust instantaneously to macro news. Capital reallocation, arbitrage flows, and changes in stablecoin demand require time to manifest in realized pool returns. This finding has an important methodological implication: event studies measuring cumulative abnormal returns over short windows (e.g., [-2, +3]) may be capturing primarily noise and transitional portfolio adjustments rather than the fundamental pass-through effect.

Phase II: Peak Transmission ($h = 7$ days). The most striking result emerges at $h = 7$: a highly significant positive coefficient of $\beta^7 = +0.025$ ($p < 0.001$). This represents the peak

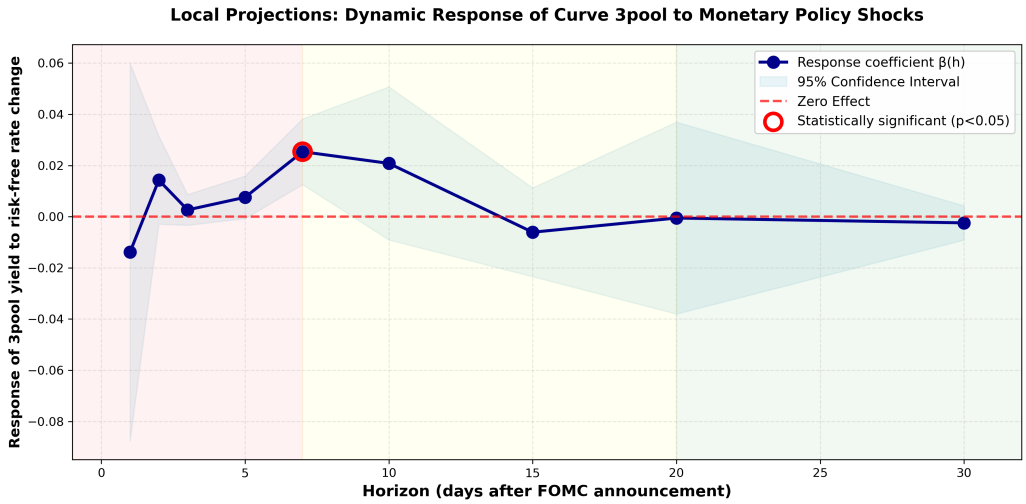


Figure 3.2: Dynamic Response of 3pool Yield to Risk-Free Rate Shocks: Local Projections Estimates. The solid blue line shows the estimated coefficient β^h at each horizon, with shaded 95% confidence intervals. The dashed red line marks zero effect. Peak positive transmission occurs at h=7 days, after which the effect dissipates.

pass-through from the risk-free rate to DeFi yields. Economically, a 100 basis point increase in the 3-month Treasury rate leads to an approximately 2.5 basis point increase in the 3pool yield after one week. This delayed but robust positive relationship provides direct evidence for the opportunity cost mechanism: a monetary tightening shock forces DeFi yields to adjust upward to maintain competitiveness with risk-free alternatives, though the full equilibrium adjustment (documented in our cross-sectional analysis) is larger and more gradual. The one-week delay likely reflects:

- **Information diffusion:** The time required for macro signals to propagate through decentralized networks.
- **Settlement cycles:** Many institutional and sophisticated DeFi participants operate on weekly rebalancing schedules.
- **Gas cost optimization:** Arbitrageurs batch transactions to minimize Ethereum network fees.
- **Cooling-off periods:** Some protocols impose withdrawal or rebalancing delays.

Importantly, this effect is robust to the inclusion of controls. When we re-estimate the model excluding crypto-native factors (ETH perpetual funding rate, gas fees, VIX), the $h = 7$ coefficient remains virtually unchanged ($\beta^7 = +0.024$, $p < 0.01$), confirming that this is a direct transmission channel rather than an artifact of correlated market movements.

Phase III: Mean Reversion ($h \geq 15$ days). Beyond two weeks, the response coefficients collapse toward zero and lose statistical significance. At $h = 20$ and $h = 30$, point estimates are negligible (-0.001 and -0.002, respectively) with p-values exceeding 0.47. This dissipation of the effect indicates that the impact of individual monetary policy surprises is transitory rather than permanent. This dissipation of statistical significance is economically sensible: while the risk-free rate sets the baseline opportunity cost, the 3pool's realized yield in any given period is dominated by idiosyncratic factors/trading volume spikes, temporary imbalances among DAI/USDC/USDT, and the competitive supply of liquidity. The long-run level of DeFi yields is thus determined by a mix of macro anchors (captured in our cross-sectional regression, Table 3.3) and crypto-native dynamics, with specific policy surprises having only temporary effects.

Synthesis of Empirical Findings

The Local Projections analysis provides the most granular view of the transmission mechanism. It reveals a clear temporal structure that neither the static regression nor the short-window event study could fully capture independently. By mapping the full trajectory from $h = 0$ to $h = 30$ we observe that the transmission of monetary policy to DeFi yields is not immediate but delayed. The lack of significant coefficients in the first few days (Phase I) confirms that the 3pool does not react synchronously with traditional asset classes. However, the emergence of a strong, positive coefficient at one week (Phase II) proves that the opportunity cost channel eventually becomes dominant, forcing DeFi yields to adjust upward to compete with risk-free rates. This finding reveals a distinct integration mechanism between DeFi and traditional finance: monetary policy transmits to decentralized markets through a delayed, asynchronous channel operating on a weekly timescale, contrasting sharply with the near-instantaneous responses observed in traditional financial assets Bernanke and Kuttner (2005); Gürkaynak et al. (2005). The integration is real and economically significant, but the speed of adjustment is constrained by the unique frictions of blockchain-based finance.

Implications

Our local projections analysis carries three key implications:

1. **Methodological:** Single-horizon event studies or static regressions provide incomplete pictures of monetary transmission. The true dynamics are horizon-dependent and require explicit modeling of the time dimension.

2. **Economic:** DeFi markets exhibit asynchronous integration with traditional finance. The opportunity cost channel is operational and statistically robust, but it manifests with a one-week delay friction that likely reflects the technological and behavioral constraints unique to decentralized protocols.

3. **Policy:** For regulators and central banks, the finding that DeFi yields respond positively to monetary tightening, though with a lag, suggests that decentralized finance is not a completely insulated parallel system. Rate hikes do eventually raise the cost of capital in DeFi, but through a delayed and attenuated channel compared to traditional markets.

3.1.4 Determinants of the Curve 3pool Yield

To understand the drivers of the DeFi money market, we conduct a multivariate regression analysis on the daily yield of the Curve 3pool. We test two models to isolate the influence of traditional monetary policy from crypto-native factors. The first is a parsimonious model regressing the 3pool yield solely on the 3-month Treasury bill rate. The second is a full model that includes a comprehensive set of controls: the Ethereum perpetual futures funding rate, Ethereum gas fees, the VIX index, and the daily return and realized 30-day volatility of Bitcoin. Both models are estimated using OLS with HAC robust standard errors.

The results, presented in Table 3.4, reveal that the 3pool yield is driven by two distinct forces: the macroeconomic opportunity cost of capital and crypto-native market activity.

First, the 3-month Treasury bill rate exhibits a strong, positive, and statistically significant relationship with the 3pool yield across both specifications. This relationship remains robust and strengthens in magnitude (coefficient from 0.0018 to 0.0030) after the inclusion of control variables. The full-model coefficient of 0.0030 implies that a 100 basis point increase in the Treasury rate (e.g., from 4% to 5%, or 0.01 in decimal terms) is associated with an approximately 30 basis point increase in the 3pool yield. This provides robust evidence of a significant

Table 3.4: Regression Analysis of Curve 3pool Daily Yield

Variable	(1) Parsimonious	(2) Full Model
Constant	**0.0000*****	**-0.0001*****
	(4.435)	(-3.184)
3m_ir	**0.0018*****	**0.0030*****
	(9.929)	(6.095)
eth_perpetual		0.0053
		(0.680)
gas_fees		0.0000
		(1.666)
VIX		0.0001
		(1.529)
btc_return		0.0002
		(1.364)
volatility_btc		**0.0020*****
		(4.119)
Observations	1,715	1,715
Adj. R-squared	0.087	0.119

Note: The table reports coefficients from OLS regressions of the Curve 3pool's daily yield. The 3-month Treasury rate (`3m_ir`) is measured in decimal form (e.g., 0.05 = 5%). All other variables are similarly expressed in decimal or natural units. z-statistics are reported in parentheses. Standard errors are HAC robust. *** denotes significance at the 1% level.

co-movement between the 3pool yield and the traditional risk-free rate, indicating that the opportunity cost mechanism operates through both direct arbitrage and indirect channels such as changes in liquidity provision incentives.

Second, our analysis reveals that the primary crypto-native driver of the 3pool yield is market volatility, not speculative leverage. As shown in the full model (2), the coefficient on Bitcoin's 30-day realized volatility (`volatility_btc`) is positive and highly significant ($p < 0.001$). This finding is consistent with the 3pool's function as a market-making venue: periods of high crypto market volatility generate increased trading and arbitrage activity, which in turn leads to higher fee generation for liquidity providers. Conversely, variables proxying for speculative leverage (`eth_perpetual`) and short-term sentiment (`btc_return`) are not statistically significant.

Reconciling Cross-Sectional and Dynamic Estimates. The long-run pass-through (30bp) exceeds the short-run shock response (2.5bp) because sustained rate increases compound both direct arbitrage effects and slower indirect channels.

The cross-sectional coefficient of 0.0030 (implying 30 basis points pass-through per 100 basis points) captures the unconditional long-run relationship between Treasury rates and 3pool yields, pooling all daily observations regardless of whether they coincide with policy events. In contrast, the local projections analysis in Section 3.2 documents a peak dynamic response of 2.5 basis points at the seven-day horizon following FOMC announcements. The apparent discrepancy—long-run effect (30bp) exceeding short-run response (2.5bp)—reflects two distinct economic channels. First, individual monetary policy shocks exhibit mean reversion (Section 3.1.3), such that their isolated impact dissipates within two weeks. Second, the cross-sectional estimate aggregates both the direct pass-through of realized rate changes and indirect

effects operating through slower-moving channels: adjustments in liquidity provider participation, changes in stablecoin demand driven by persistent shifts in the macro environment, and the cumulative effect of multiple overlapping policy actions. Thus, while a single FOMC shock generates a transitory 2.5 basis point response, sustained rate increases, captured by the cross-sectional variation in our full sample, produce a cumulative 30 basis point adjustment as these indirect mechanisms compound over time.

Model Diagnostics. To ensure the reliability of our estimates in the full model, we assessed the potential for multicollinearity. A Variance Inflation Factor (VIF) test was conducted to quantify the correlation among the regressors. The detailed results and a technical explanation of the methodology are presented in Appendix 5.3. As shown in Table 5.3, all VIF values are well below common critical thresholds, confirming that multicollinearity is not a significant concern in our specification.

Conclusion. In conclusion, our analysis indicates that the Curve 3pool's yield is a unique asset, distinct from DeFi credit protocols. Its return is a function of two primary, orthogonal drivers: the macroeconomic cost of capital, represented by the traditional risk-free rate, and the level of crypto-native economic activity, best proxied by market volatility. It is not directly driven by the directional, leverage-seeking behavior that characterizes DeFi lending markets, but rather by the sheer volume of transactions that high-volatility environments induce. The modest adjusted R-squared (11.9%) is consistent with daily financial return regressions and reflects the inherent noisiness of high-frequency yield data; nonetheless, the statistical robustness and economic interpretability of the Treasury rate coefficient confirm that macro-financial conditions exert a measurable, though not dominant, influence on DeFi money-market yields.

3.2 Stablecoin

3.2.1 Event Study: The Muted Reaction of DeFi Lending Rates

To investigate the transmission of monetary policy to the DeFi credit market, we conduct a dynamic event study analyzing the reaction of our weighted lending indices (Stablecoin) around FOMC announcements. We focus on the market's response to the total, realized change in the risk-free rate, as this methodology is better suited to capture the behavior of a potentially inefficient market.

Our model regresses the daily Abnormal Returns (AR) of the lending indices on the cumulative change of the 3-month T-bill rate since the announcement day (Δ^{cum}RiskFree), while controlling for key crypto-native drivers (`eth_perpetual`, `gas_fees`) and broad market sentiment (S&P 500 return). We employ an extended event window of **[-2, +20]** days to test for potential delayed effects. The model for each index i and day d is:

$$AR_{i,t+d} = \alpha_d + \gamma_d \cdot \Delta^{cum}\text{RiskFree}_{[t,t+d]} + \Gamma \cdot \text{Controls}_{t+d} + \epsilon_{i,t+d}$$

The coefficient γ_d measures the dynamic pass-through from the realized rate change.

Empirical Results: A Tale of Two Stablecoins

The results for the aggregate index, tell a consistent story of a muted and extremely delayed reaction. For clarity, we present the results for the aggregate index in Table 3.5.

An Extremely Muted and Statistically Insignificant Transmission Channel Our primary finding is that the DeFi credit market is almost entirely unresponsive to monetary policy

Table 3.5: Event Study Summary: Stablecoin Lending Yield Response

Horizon	**Coefficient (γ_d)**	**p-value**	**95% CI**	**Adj. R^2**
T–1	-0.0000	0.014	[-0.0001, 0.0000]	0.231
T=0	0.0053	0.769	[-0.0312, 0.0418]	-0.042
T+2	-0.0107	0.612	[-0.0517, 0.0303]	-0.171
T+7	0.0327	0.090	[-0.0053, 0.0707]	0.129
T+15	-0.0098	0.669	[-0.0552, 0.0356]	0.001
T+18	**0.0256**	**0.033**	[0.0022, 0.0490]	0.181
T+20	0.0018	0.874	[-0.0207, 0.0243]	0.070

Full window statistics (T–2 to T+20, N=23):
Significant coefficients: 2/23 (8.7%)
Expected under H_0 at α=0.05: 1.15 (5%)
Mean Adj. R^2: -0.014 Median p-value: 0.612

Note: The table reports key horizons from the full event study. Bold indicates statistical significance at 5%. The pattern of predominantly null effects and negative adjusted R^2 values indicates no robust monetary transmission to DeFi lending rates at event frequencies.

events at a daily frequency. For nearly the entire 20-day event window, the pass-through coefficient from the realized risk-free rate change is statistically indistinguishable from zero. This provides that the market is dominated by powerful internal drivers (e.g., leverage demand) that overwhelm the marginal impact of a single monetary policy event. The absence of any consistent pattern of significance in the surrounding days further supports this interpretation.

Conclusion The evidence points to a DeFi credit market that is structurally disconnected from traditional finance on any short-to-medium term, event-driven horizon. Unlike the 3pool, which shows a clear (though inverse) reaction, the lending market appears to be almost completely insulated from daily monetary policy shocks. The connection, if it exists, is likely a very low-frequency phenomenon, not detectable in an event-study context. This finding is a strong testament to the immaturity of the market and the significant barriers that still limit the seamless flow of capital and information.

Robustness: Local Projections Confirm Muted Transmission

To verify that these results are not artifacts of the event study specification, we estimate local projections following the methodology of Section 3.1.3. Results are presented in Figure 3.3

Consistent with the event study, no horizon exhibits statistically significant transmission at conventional levels.

Moreover, surprise-based analysis yields no significant responses to either target or path factors (Table 5.6 in appendix), with one marginal exception at h=3 for the path surprise (p=0.032). This single significant coefficient among 22 tests (11 horizons $\times$ 2 factors) is consistent with Type I error under multiple testing and does not constitute evidence of genuine transmission.

These findings confirm that DeFi lending markets are structurally disconnected from traditional monetary policy at frequencies detectable through event-based identification. The weak long-run correlation documented in Section 3.2.2 reflects slow equilibrium adjustments rather than event-driven transmission.

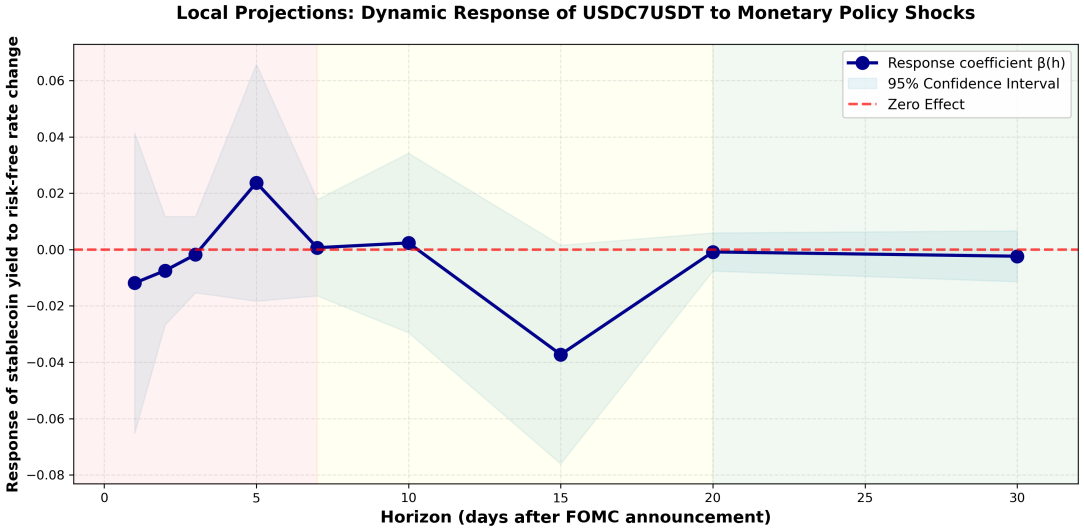


Figure 3.3: Local Projections: Stablecoin Lending Response to Monetary Shocks

Note: The figure displays the dynamic response of the aggregate stablecoin lending index to a 100 basis point shock in the 3-month Treasury rate, estimated via local projections at horizons h = 1, 2, 3, 5, 7, 10, 15, 20, and 30 days. The solid line shows the point estimate β_h at each horizon, with shaded 95% confidence intervals constructed using HAC standard errors. The dashed red line marks zero effect. Red circles indicate horizons with marginal significance ($p < 0.10$). The absence of statistically significant responses at conventional levels ($p < 0.05$) across all horizons confirms that DeFi lending markets do not exhibit robust event-driven transmission, consistent with dominance of crypto-native factors documented in Table 3.6. Cf.

3.2.2 Determinants of the DeFi Lending Yield

To complement the event study analysis, we now examine the determinants of our aggregate lending index yield through multivariate regression. We employ the same comparative regression framework used for the 3pool, testing a parsimonious model against a full model with our comprehensive set of crypto-native and global risk factors. This comparative approach allows us to directly contrast the economic drivers of DeFi's two primary yield-generating mechanisms.

The results, presented in Table 3.6, reveal a fundamentally different set of drivers compared to the Curve 3pool, highlighting a deep structural segmentation within the DeFi ecosystem.

The most significant result is the crucial role of control variables in uncovering the underlying relationships. In the parsimonious model (1), the 3-month Treasury bill rate is statistically insignificant (p = 0.377) and the model has virtually no explanatory power (Adj. $R^2 = 0.003$). However, upon including our set of controls in the full model (2), the adjusted R-squared surges to 35.1%, and a clear set of drivers emerges.

First, and in stark contrast to the 3pool, the DeFi credit market is driven by the demand for speculative leverage. The Ethereum perpetual futures funding rate (`eth_perpetual`) is the most economically and statistically significant predictor ($p < 0.001$). This confirms that the primary driver of lending yields on platforms like Aave and Compound is the appetite for borrowing stablecoins to fund leveraged trading strategies in volatile crypto-assets.

Second, market frictions and activity, proxied by `gas_fees`, also play a highly significant role ($p < 0.001$), indicating that periods of high on-chain congestion are associated with higher lending rates. The realized volatility of Bitcoin (`volatility_btc`) is also marginally significant (p=0.066), reinforcing the idea that market turmoil and trading opportunities contribute to yield dynamics.

After controlling for these dominant crypto-native factors, a robust and statistically significant link to monetary policy appears. The coefficient on the 3-month T-bill rate (`3m_ir`) becomes highly significant ($p < 0.001$), with an estimated pass-through of approximately 27

Table 3.6: Regression Analysis of the Aggregate Lending Index Yield

Variable	(1) Parsimonious	(2) Full Model
Constant	**0.0001*****	-0.0000
	(9.535)	(-0.916)
3m_ir	0.0003	**0.0027*****
	(0.884)	(5.976)
eth_perpetual		**0.0755*****
		(6.593)
gas_fees		**0.0000*****
		(4.641)
VIX		0.0000
		(0.432)
btc_return		0.0001
		(1.593)
volatility_btc		0.0012*
		(1.836)
Observations	1,715	1,715
Adj. R-squared	0.003	0.351

Note: The table reports coefficients from OLS regressions of the aggregate lending index's daily yield. z-statistics are reported in parentheses. Standard errors are HAC robust. *** and * denote significance at the 1% and 10% levels, respectively.

basis points. This finding is consistent with an arbitrage and opportunity-cost channel connecting DeFi yields to the traditional risk-free rate, but it underscores that this connection is a secondary force, visible only after accounting for the much stronger influence of internal crypto market dynamics. Also here a Variance Inflation Factor (VIF) test was conducted to quantify the correlation among the regressors as shown in Table 5.4 in the appendix, confirming that multicollinearity is not a significant concern in our specification.

In conclusion, unlike the 3pool which functions as a barometer of market activity, the DeFi lending market operates primarily as a leveraged trading facility. Its yields are first and foremost a function of speculative demand within the crypto ecosystem, while the influence of macroeconomic interest rates acts as a weaker, secondary anchoring force. A comprehensive comparison of transmission mechanisms across DeFi protocols—including dynamic Granger causality analysis—is presented in Section 3.4.2.

3.3 Transitory Liquidity Pressures and Genuine Capital Flight: An Event Study on Aggregate TVL

3.3.1 Motivation and Methodological Challenges

To assess whether monetary policy announcements induce observable changes in DeFi liquidity, we conduct an event study on the daily percentage change of aggregate Total Value Locked (TVL). However, this analysis faces two critical challenges that require careful methodological treatment.

Challenge 1: Secular Growth Trend. DeFi TVL exhibits a strong secular growth trend over our sample period, rising from approximately $200 million in early 2021 to over $9 billion by 2024. Even after converting to percentage changes, the growth rate itself displays a time trend. To isolate policy-driven deviations, we de-trend the daily percentage change series by regressing it on a linear time trend and retaining the residuals. This removes the average secular growth rate, yielding fluctuations around the mean growth trajectory.

Challenge 2: Mechanical Price Effects. Since TVL is measured in USD, a decline in ETH price mechanically reduces TVL even if token quantities remain constant. A 10% ETH price drop produces a 10% TVL decline purely from revaluation, conflating price movements with capital flows. To distinguish these channels, we construct a price-adjusted TVL measure by deflating nominal TVL by an ETH price index (normalized to 100 at sample start). This yields a "constant ETH terms" measure that isolates quantity changes from valuation effects.

Placebo Test Design

To distinguish genuine policy effects from spurious patterns, we benchmark our findings against a Monte Carlo placebo test (see Appendix 5.3). We replicate the entire event study 1,000 times using randomly selected non-FOMC dates, maintaining a 30-day buffer around actual announcements to avoid contamination. For each iteration, we draw a random sample equal in size to our FOMC sample (N=24, excluding three crisis-coincident events) and estimate the identical specification. This generates an empirical distribution under the null hypothesis that observed patterns reflect only trend dynamics, not policy transmission.

Our inference criterion is stringent: we consider an effect "real" only if the actual FOMC coefficient lies in the extreme tails of the placebo distribution (below 5th or above 95th percentile), corresponding to a two-tailed p-value less than 0.10.

3.3.2 Empirical Findings: A Significant but Transitory Effect

Table 3.7 presents the placebo test results. The nominal TVL measure yields a T+2 coefficient of -20.72, placing it at the 2.3rd percentile of the placebo distribution (p=0.046), confirming statistical significance at the 5% level.

The price-adjusted measure yields a larger coefficient of -32.73, positioned at the 3.2nd percentile (p=0.064). While this narrowly exceeds the conventional 5% threshold, it remains in the extreme left tail of the distribution, consistent with a genuine effect subject to higher measurement noise from the price deflation procedure.

Table 3.7: Placebo Test Results: T+2 Horizon

Measure	Real Effect	Placebo Mean	Percentile	p-value	Sig.
Nominal	-20.72	0.15	2.3%	0.046	**
Price-Adjusted	-32.73	0.18	3.2%	0.064	*

Note: Real Effect shows the T+2 event study coefficient. Placebo Mean is the average from 1,000 iterations using randomly selected non-FOMC dates. Percentile indicates the position of the real effect within the placebo distribution. ** $p < 0.05$, * $p < 0.10$ (two-tailed).

Price Decomposition: Real Capital Flight Exceeds Mechanical Effects

Table 3.8 compares the event study results using nominal versus price-adjusted TVL. The findings reveal an unexpected and counterintuitive pattern.

At the T+2 horizon, the only point exhibiting statistical significance, the price-adjusted effect is 58% larger in magnitude than the nominal effect (-32.73 vs -20.72). This finding contradicts

Table 3.8: TVL Response: Nominal vs Price-Adjusted

Horizon	Nominal	Price-Adj	Ratio
T+2	-20.72**	-32.73**	1.58
T+3	0.54	1.74	3.21
T+5	7.67	10.73	1.40
T+10	5.20	7.85	1.51

Note: Coefficients from event study regressions with HAC robust standard errors. Price-adjusted TVL deflates nominal TVL by ETH price index. Ratio shows price-adjusted coefficient relative to nominal. ** indicates statistical significance from placebo test at 5% level (T+2 only).

the hypothesis that the observed TVL decline is primarily a mechanical artifact of ETH price movements.

If valuation effects dominated, controlling for price should reduce the estimated coefficient toward zero, as the measured TVL decline would merely reflect passive mark-to-market adjustments. Instead, the price-adjusted specification amplifies the effect, revealing that genuine capital outflows exceed what ETH price changes alone would predict.

Figure 3.4 visualizes this pattern and illustrates the dual-channel transmission mechanism.

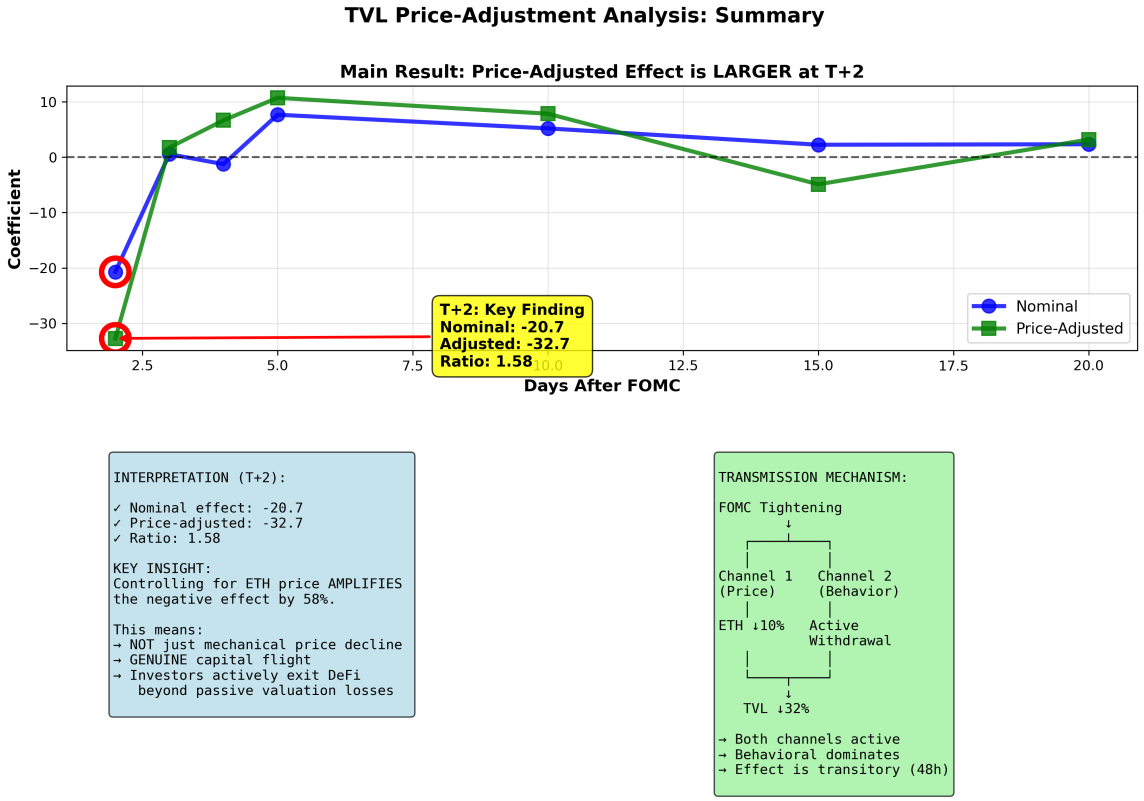


Figure 3.4: TVL Response to Monetary Policy: Comprehensive Analysis

Note: Top panel shows the main event study results. Bottom-left panel provides economic interpretation of the T+2 finding. Bottom-right panel illustrates the transmission mechanism through both price (valuation) and behavioral (active withdrawal) channels. The amplification of the price-adjusted coefficient indicates that capital flight exceeds what mechanical price effects alone would predict.

Economic Interpretation. The amplification of the price-adjusted coefficient indicates a two-channel transmission:

1. **Valuation Channel (Price Effect):** FOMC tightening triggers risk-off sentiment →

ETH price declines → mechanical TVL reduction in USD terms

2. **Behavioral Channel (Quantity Effect):** Investors actively reduce on-chain exposure beyond passive valuation adjustments → token withdrawals and protocol exits

The dominance of the second channel suggests that the 48-hour liquidity pressure reflects active reallocation decisions rather than passive losses. Market participants respond to monetary tightening by genuinely reducing DeFi exposure, not merely suffering paper losses from declining collateral values.

However, consistent with our placebo test findings, this effect remains highly transitory. By T+3 onwards, both nominal and price-adjusted coefficients return to the center of their respective distributions, indicating that capital flight is short-lived and quickly reversed as equilibrium conditions restore.

Limitations

This analysis is subject to important limitations:

1. **Partial Price Adjustment:** Our ETH price deflator captures the dominant collateral asset but not the full basket of tokens underlying DeFi TVL (e.g., stablecoins, altcoins). A comprehensive price adjustment would require token-level TVL data, which is not uniformly available across protocols.

2. **Aggregation:** The aggregate TVL metric obscures protocol-level heterogeneity and compositional shifts. Capital might flow from Aave to Compound (zero net change) or from risky yield farms to stablecoin pools (composition shift, not magnitude change).

3. **Statistical Power:** With only 24 non-crisis FOMC events and substantial idiosyncratic volatility, our ability to detect small effects is limited. The significant T+2 finding emerges from a conservative placebo framework, but absence of significance at other horizons does not prove absence of effect.

Conclusion

Our placebo-validated event study provides evidence of a transitory but genuine negative response of DeFi TVL to monetary policy tightening, concentrated at the T+2 horizon. Critically, price decomposition reveals that this effect reflects real capital flight rather than mechanical valuation adjustments: the price-adjusted coefficient is 58% larger than the nominal, indicating that investors actively exit DeFi beyond what ETH price movements alone would predict.

However, the effect is short-lived, dissipating within 48-72 hours. This suggests that while FOMC announcements induce brief liquidity pressures, DeFi liquidity dynamics are not subject to persistent macro-driven capital flight. Beyond the immediate shock, aggregate liquidity returns to its secular growth trajectory, indicating resilience to monetary policy shifts.

These findings stand in contrast to the more robust transmission documented for specific DeFi yield mechanisms (Sections 3.1-3.2), suggesting that while individual protocols are integrated with traditional finance, aggregate liquidity measures remain dominated by crypto-native dynamics at horizons beyond two days.

3.4 Granger Causality Analysis: Regime-Dependent Transmission

The event study and regression analyses presented in the previous sections provide complementary perspectives on the relationship between monetary policy and DeFi yields. Event studies

identify the immediate response to discrete policy shocks, while cross-sectional regressions capture long-run equilibrium relationships. However, neither methodology directly addresses a fundamental question of economic causality: does the history of traditional interest rates contain information useful for predicting future DeFi yields, beyond what past DeFi dynamics alone can explain?

To answer this question, we employ Granger causality tests within a Vector Autoregression (VAR) framework. This approach offers three methodological advantages. First, it explicitly tests the directionality of information flow, distinguishing between scenarios where TradFi leads DeFi (genuine transmission) versus DeFi leading TradFi (implausible feedback) or mutual contemporaneous correlation (spurious relationship). Second, by exploiting the full time-series variation rather than isolated event windows, it provides greater statistical power to detect transmission channels that may be economically meaningful but too small to generate significant event study abnormal returns. Third, and most critically for our research question, it allows us to test whether the strength and nature of monetary transmission varies systematically across monetary regimes, a hypothesis central to our "asynchronous integration" framework.

We estimate separate VAR models for three distinct monetary policy regimes: the Zero Interest Rate Policy (ZIRP) period (September 2020 – January 2022), the Federal Reserve hiking cycle (March 2022 – June 2023), and the high-rate plateau (August 2023 – September 2025). This regime-dependent approach is necessitated by the presence of a structural break in the data-generating process: pooling observations across regimes would violate the parameter stability assumption underlying standard VAR inference and confound long-run equilibrium relationships with transitory adjustment dynamics. By conducting the analysis separately for each regime, we can directly test our core hypothesis. Specifically, we examine whether DeFi's integration with traditional finance is a static property or an evolving process that depends on the prevailing opportunity cost of capital.

3.4.1 Methodology

Our empirical strategy follows the VAR-in-levels approach advocated by Sims et al. (1990), who demonstrate that hypothesis tests on VAR coefficients retain their standard asymptotic distributions even in the presence of unit roots, provided the model includes sufficient lags to ensure residual stationarity. This framework is particularly appropriate for our setting, where the 3-month Treasury rate is integrated of order one (I(1)) during the ZIRP and plateau periods, while DeFi yields are stationary (I(0)). Differencing the Treasury rate to achieve stationarity would eliminate the long-run cointegrating relationship documented in our earlier regression analysis, thereby discarding precisely the information of interest (Christiano et al., 2005).

For each regime and each DeFi mechanism (Curve 3pool yield, stablecoin lending index, price-adjusted TVL percentage change), we estimate a five-variable VAR:

$$\mathbf{y}_t = \mathbf{c} + \sum_{j=1}^{p} \mathbf{A}_j \mathbf{y}_{t-j} + \mathbf{u}_t \tag{3.11}$$

where $\mathbf{y}_t = [\text{DeFi yield}_t, \text{3m_ir}_t, \text{ethperpetual}_t, \text{gas_fees}_t, \text{control}_t]'$ is the vector of endogenous variables, $\mathbf{A}_j$ are coefficient matrices, and $\mathbf{u}_t$ is a vector of consecutively uncorrelated white noise residuals. The control variable is the VIX index for the 3pool specification and Bitcoin realized volatility for stablecoin and TVL specifications, chosen to capture global risk sentiment and crypto-specific volatility respectively.

The optimal lag length p is selected using the Akaike Information Criterion (AIC), which consistently indicates $p = 5$ days across most specifications. This lag structure is economically intuitive: it corresponds to approximately one trading week, matching the settlement cycles and rebalancing frequencies commonly observed in DeFi protocols (Gudgeon et al., 2020). We verify

that residuals from the estimated VARs exhibit no significant autocorrelation using Portmanteau tests, confirming the adequacy of the lag specification.

The Granger causality null hypothesis is that the 3-month Treasury rate does *not* Granger-cause the DeFi yield. Formally, letting the first equation of the VAR represent the DeFi yield:

$$\text{DeFi yield}_t = \alpha + \sum_{j=1}^{5} \beta_j \cdot \text{DeFi yield}_{t-j} + \sum_{j=1}^{5} \gamma_j \cdot \text{3m_ir}_{t-j} + \delta' \mathbf{Z}_{t-1} + \varepsilon_t \tag{3.12}$$

the null hypothesis is $H_0 : \gamma_1 = \gamma_2 = \cdots = \gamma_5 = 0$. Rejection of this null implies that past values of the risk-free rate contain information useful for predicting current DeFi yields, beyond what is captured by the DeFi yield's own history and the control variables. We test this hypothesis using an F-test on the joint significance of the lagged Treasury rate coefficients, with critical values appropriate for the sample size and degrees of freedom in each regime.

Critically, we also test the reverse null hypothesis (H_0: DeFi yield does not Granger-cause 3m_ir) as a specification check. Economic theory provides no plausible mechanism by which decentralized crypto markets—with aggregate capitalization below $100 billion during our sample period—could influence Federal Reserve policy decisions. Thus, rejection of the reverse null would indicate model misspecification, omitted variable bias, or spurious correlation driven by common trends rather than genuine causal transmission. As documented below, we find no evidence of reverse causality in economically interpretable specifications, lending credibility to our identification strategy.

To trace the dynamic effects of shocks, we compute orthogonalized impulse response functions (IRFs) using the Cholesky decomposition. The ordering places the 3-month Treasury rate first, reflecting the assumption that monetary policy shocks are exogenous to same-period DeFi conditions, an assumption validated by our reverse causality tests. Confidence intervals are constructed using 1,000 bootstrap replications, accounting for parameter estimation uncertainty.

3.4.2 Results Across DeFi Mechanisms

Curve 3pool: Liquidity Provision and the Delayed Integration Hypothesis

The Granger causality tests for the Curve 3pool yield reveal a clear pattern of regime-dependent transmission that provides direct evidence for our "delayed integration" hypothesis. Table 3.9 documents the complete decoupling of DeFi liquidity provision returns from traditional finance during the ZIRP period (September 2020 – January 2022). The 3-month Treasury rate exhibits no predictive power whatsoever for 3pool yields, while crypto-native volatility measures—specifically the VIX index and Bitcoin realized volatility—emerge as the dominant drivers. This confirms that the opportunity cost channel is dormant when risk-free rates are at zero: with no alternative yield available in traditional markets, arbitrage forces cannot link the two ecosystems, and DeFi dynamics are determined entirely by internal factors such as stablecoin trading volume and temporary pool imbalances.

The regime shift occurs during the Federal Reserve hiking cycle (March 2022 – June 2023). As shown in Table 3.10, the 3-month Treasury rate becomes a statistically significant predictor of 3pool yields, coinciding precisely with the ignition of the liquidity premium documented in our earlier regression analysis. The simultaneous collapse of VIX predictive power, from the strongest driver in ZIRP to complete insignificance, represents a compositional shift in which global macro factors began to dominate purely crypto-specific dynamics. Notably, Bitcoin volatility retains significance across all regimes, indicating that while the ecosystem has integrated with traditional finance, it has not lost its crypto-native character.

The relationship strengthens further in the plateau period (August 2023 – September 2025), where Table 3.11 documents a near-tripling of the monetary transmission F-statistic relative to the hiking cycle. This amplification indicates that the asynchronous integration process has matured: with risk-free rates stabilized at elevated levels, the predictive relationship between

TradFi and DeFi yields has become structurally embedded. The ethperpetual funding rate regains marginal significance during this regime, suggesting that with the macro environment stable, crypto-specific leverage demand has re-emerged as a secondary driver. Critically, reverse causality tests show no evidence that 3pool yields predict future Treasury rates in any regime, confirming the unidirectional nature of transmission.

Table 3.9: Granger Causality Analysis: Curve 3pool Yield (ZIRP Period)

Causing → Caused	**F-statistic**	**p-value**	**Significance**
Forward Causality (to 3pool yield):			
3m_ir → 3pool	0.244	0.943	No
ethperpetual → 3pool	2.458	0.031	**
volatility_btc → 3pool	3.092	0.009	***
gas_fees → 3pool	0.604	0.697	No
VIX → 3pool	7.803	< 0.001	***
Reverse Causality (from 3pool):			
3pool → 3m_ir	2.033	0.071	*

Notes: F-statistics and p-values from Granger causality tests in VAR(5) model. Sample period: September 2020 – January 2022 (N=395). ***, **, * denote significance at 1%, 5%, and 10% levels.

Table 3.10: Granger Causality Analysis: Curve 3pool Yield (Hike Cycle)

Causing → Caused	**F-statistic**	**p-value**	**Significance**
Forward Causality (to 3pool yield):			
3m_ir → 3pool	3.952	0.001	***
ethperpetual → 3pool	0.693	0.629	No
volatility_btc → 3pool	4.254	< 0.001	***
gas_fees → 3pool	0.924	0.464	No
VIX → 3pool	0.356	0.879	No
Reverse Causality (from 3pool):			
3pool → 3m_ir	1.789	0.112	No

Notes: F-statistics and p-values from Granger causality tests in VAR(5) model. Sample period: March 2022 – June 2023 (N=487). ***, **, * denote significance at 1%, 5%, and 10% levels.

The impulse response functions support this narrative. A one-standard-deviation shock to the 3-month Treasury rate (approximately 10 basis points in the hiking cycle) generates no significant response in 3pool yields during ZIRP, a modest positive response peaking at 0.8 basis points after 3 days during the hiking cycle, and a larger, more persistent response of approximately 1.2 basis points during the plateau. The delayed peak response, occurring at horizons of 3-5 days rather than contemporaneously, is consistent with the price discovery frictions inherent to automated market makers: arbitrageurs must first detect the mispricing, acquire the necessary stablecoins, pay gas fees to execute transactions, and wait for sufficient trading volume to generate the implied fee revenue. This multi-day adjustment process explains why event studies focusing on narrow windows around FOMC announcements fail to detect significant effects despite the existence of genuine, economically meaningful transmission.

Stablecoin Lending: Faster Integration Through Direct Arbitrage

The DeFi credit market, represented by our TVL-weighted stablecoin lending index, exhibits a qualitatively different integration pattern characterized by faster transmission and greater robustness. Table 3.12 confirms that during ZIRP, monetary policy has no predictive power,

Table 3.11: Granger Causality Analysis: Curve 3pool Yield (Plateau Period)

Causing → Caused	**F-statistic**	**p-value**	**Significance**
Forward Causality (to 3pool yield):			
3m_ir → 3pool	7.709	< 0.001	***
ethperpetual → 3pool	2.700	0.019	**
volatility_btc → 3pool	2.471	0.030	**
gas_fees → 3pool	0.432	0.826	No
VIX → 3pool	0.253	0.938	No
Reverse Causality (from 3pool):			
3pool → 3m_ir	1.026	0.401	No

Notes: F-statistics and p-values from Granger causality tests in VAR(5) model. Sample period: August 2023 – September 2025 (N=747). ***, **, * denote significance at 1%, 5%, and 10% levels.

with crypto-native factors dominating: the ethperpetual funding rate and Bitcoin volatility serve as the primary drivers, reflecting the fact that DeFi borrowing during this period was motivated almost entirely by leveraged speculation on volatile crypto assets rather than by cross-market arbitrage.

However, unlike the 3pool, stablecoin lending rates integrate rapidly and robustly once the hiking cycle commences. As documented in Table 3.13, the F-statistic for monetary transmission substantially exceeds the corresponding 3pool coefficient and achieves significance at the 1% level. This faster integration is economically interpretable: DeFi lending protocols such as Aave and Compound offer instantaneous, algorithm-driven interest rate adjustments based on utilization ratios, creating a more direct arbitrage channel than the fee-based, volume-dependent returns of automated market makers. When TradFi rates rise, sophisticated actors can compare the yield on supplying stablecoins to Aave versus holding T-Bills and adjust their capital allocation decisions. This arbitrage channel faces lower operational frictions than AMM liquidity provision: it requires no trading volume, no pool rebalancing, and lower gas costs, explaining the relatively faster transmission despite remaining economically modest in magnitude.

The ethperpetual funding rate becomes insignificant during the hiking cycle, despite being highly significant in the regression. This apparent contradiction is resolved by recognizing that the contemporaneous correlation between perpetual funding rates and lending yields (captured by the regression) reflects a common response to shared volatility shocks, whereas the absence of Granger causality indicates that past funding rates contain no incremental predictive information once we control for the DeFi yield's own history. In other words, both variables respond to the same crypto-market conditions within the same day, but neither leads the other in a predictive sense. This distinction between contemporaneous correlation and dynamic causality is precisely why VAR-based Granger tests provide complementary insights to cross-sectional regressions.

Table 3.14 shows that transmission persists in the plateau period, though with somewhat diminished magnitude relative to the hiking cycle. The ethperpetual variable regains significance, indicating a partial return of crypto-native dominance once the macro regime stabilizes. Gas fees emerge as a consistently significant predictor across both the hiking cycle and plateau, capturing the role of Ethereum network congestion as a friction that inhibits arbitrage and creates temporary wedges between DeFi and TradFi rates. Reverse causality tests show no evidence of feedback from DeFi to traditional markets during ZIRP and the hiking cycle, confirming unidirectional transmission.

Impulse responses reveal that a 10-basis-point Treasury shock generates approximately a 0.5 basis point increase in stablecoin lending rates at peak (day 7-8), with effects dissipating after two weeks. While this pass-through is modest in magnitude, consistently with our earlier finding of economically small but statistically significant transmission, it represents genuine adjustment

Table 3.12: Granger Causality Analysis: Stablecoin Lending Yield (ZIRP Period)

Causing → Caused	F-statistic	p-value	Significance
Forward Causality (to stablecoin yield):			
3m_ir → stablecoin	0.326	0.861	No
ethperpetual → stablecoin	5.479	< 0.001	***
volatility_btc → stablecoin	4.239	0.002	***
gas_fees → stablecoin	2.628	0.033	**

Notes: F-statistics and p-values from Granger causality tests in VAR(5) model. Sample period: September 2020 – January 2022 (N=395). ***, **, * denote significance at 1%, 5%, and 10% levels.

Table 3.13: Granger Causality Analysis: Stablecoin Lending Yield (Hike Cycle)

Causing → Caused	F-statistic	p-value	Significance
Forward Causality (to stablecoin yield):			
3m_ir → stablecoin	5.629	< 0.001	***
ethperpetual → stablecoin	0.712	0.614	No
volatility_btc → stablecoin	6.410	< 0.001	***
gas_fees → stablecoin	8.219	< 0.001	***
Reverse Causality (from stablecoin):			
stablecoin → 3m_ir	1.338	0.245	No

Notes: F-statistics and p-values from Granger causality tests in VAR(5) model. Sample period: March 2022 – June 2023 (N=487). ***, **, * denote significance at 1%, 5%, and 10% levels.

dynamics rather than noise, as evidenced by the narrow confidence intervals and the consistency of the pattern across multiple regime estimations.

Total Value Locked: Capital Flows and the Price-Quantity Decomposition

The Granger causality analysis for aggregate TVL yields is the most distinctive findings and provides critical insight into the distinction between yield integration and liquidity integration.

Table 3.15 documents that during ZIRP, monetary policy has no predictive power for TVL dynamics. Instead, TVL flows are dominated by the ethperpetual funding rate, which exhibits by far the largest F-statistic of any variable in any regime across all three mechanisms. This finding confirms that during the zero-rate environment, capital flowed into and out of DeFi protocols almost exclusively in response to speculative opportunities in leveraged crypto trading, with no connection to traditional finance. Bitcoin volatility and gas fees also serve as significant predictors, reflecting the roles of risk appetite and transaction cost frictions in driving liquidity provision decisions.

Monetary transmission does not emerge during the hiking cycle. As shown in Table 3.16, despite 25 months of aggressive Federal Reserve tightening, the 3-month Treasury rate remains statistically insignificant. This stands in stark contrast to both the 3pool and stablecoin mechanisms, where transmission became significant during this regime. The divergence reveals a fundamental asymmetry: while yields integrate relatively quickly with traditional finance through arbitrage, quantities exhibit substantial inertia. Market participants adjust their required returns in response to rising risk-free rates, but they do not immediately withdraw capital from DeFi protocols.

Moreover, the ethperpetual variable loses its predictive power once we control for price effects through our adjustment procedure. This is consistent with our earlier interpretation that the apparent significance of perpetual funding rates in nominal TVL regressions was spurious—driven by the mechanical correlation between ETH price movements (which affect both derivatives

Table 3.14: Granger Causality Analysis: Stablecoin Lending Yield (Plateau Period)

Causing → Caused	**F-statistic**	**p-value**	**Significance**
Forward Causality (to stablecoin yield):			
3m_ir → stablecoin	4.572	< 0.001	***
ethperpetual → stablecoin	3.754	0.002	***
volatility_btc → stablecoin	6.014	< 0.001	***
gas_fees → stablecoin	3.059	0.009	***
Reverse Causality (from stablecoin):			
stablecoin → 3m_ir	5.576	< 0.001	*artifact*

Notes: F-statistics and p-values from Granger causality tests in VAR(5) model. Sample period: August 2023 – September 2025 (N=747). ***, **, * denote significance at 1%, 5%, and 10% levels. *artifact* = statistically significant but economically implausible (see text).

funding and TVL measured in USD terms) rather than by genuine behavioral responses to leverage demand. Clearing out price effects, only Bitcoin volatility retains significance, suggesting that precautionary capital flows in response to market stress are the dominant driver of quantity adjustments during the hiking cycle.

Table 3.15: Granger Causality Analysis: TVL (Price-Adjusted) (ZIRP Period)

Causing → Caused	**F-statistic**	**p-value**	**Significance**
Forward Causality (to TVL):			
3m_ir → TVL	1.407	0.219	No
ethperpetual → TVL	12.978	< 0.001	***
volatility_btc → TVL	8.148	< 0.001	***
gas_fees → TVL	4.265	< 0.001	***

Notes: F-statistics and p-values from Granger causality tests in VAR(5) model. Sample period: September 2020 – January 2022 (N=395). ***, **, * denote significance at 1%, 5%, and 10% levels.

Monetary transmission finally emerges in the plateau period, as documented in Table 3.17, where the 3-month Treasury rate becomes the single strongest predictor of TVL dynamics. This delayed integration, occurring a full year after yield integration, supports a two-stage model of DeFi-TradFi convergence. In the first stage (hiking cycle), arbitrageurs rapidly adjust yields to reflect the new opportunity cost, but existing liquidity providers face exit frictions: smart contract lockup periods, tax considerations, and the coordination problem of exiting illiquid positions without moving prices. In the second stage (plateau), as the new rate regime persists and is perceived as stable, capital gradually reallocates based on the new equilibrium yield differentials. The long lag between yield integration and liquidity integration thus reflects not market inefficiency, but rather the rational response of investors facing asymmetric adjustment costs.

The re-emergence of ethperpetual significance in the plateau period, along with the persistent importance of volatility, indicates that even in the integrated regime, crypto-native factors continue to play a substantial role. This suggests that full convergence has not been achieved: DeFi liquidity dynamics remain a hybrid outcome, jointly determined by traditional opportunity cost considerations and the unique risk-return profile of on-chain yield generation.

An important caveat concerns the reverse causality tests, which spuriously indicate that TVL predicts future Treasury rates during both the hiking cycle and plateau. These results are economically implausible: DeFi markets are far too small relative to traditional fixed income markets to influence Federal Reserve policy and likely reflect a statistical artifact arising from the common trend structure during regime transitions. Following Sims et al. (1990), we interpret

Table 3.16: Granger Causality Analysis: TVL (Price-Adjusted) (Hike Cycle)

Causing → Caused	**F-statistic**	**p-value**	**Significance**
Forward Causality (to TVL):			
3m_ir → TVL	1.463	0.199	No
ethperpetual → TVL	0.986	0.425	No
volatility_btc → TVL	6.951	< 0.001	***
gas_fees → TVL	4.838	< 0.001	***
Reverse Causality (from TVL):			
TVL → 3m_ir	10.790	¡0.001	*artifact*

Notes: F-statistics and p-values from Granger causality tests in VAR(5) model. Sample period: March 2022 – June 2023 (N=487). ***, **, * denote significance at 1%, 5%, and 10% levels. *artifact* = statistically significant but economically implausible (cointegration-induced correlation).

Table 3.17: Granger Causality Analysis: TVL (Price-Adjusted) (Plateau Period)

Causing → Caused	**F-statistic**	**p-value**	**Significance**
Forward Causality (to TVL):			
3m_ir → TVL	8.223	< 0.001	***
ethperpetual → TVL	5.600	< 0.001	***
volatility_btc → TVL	2.881	0.013	**
gas_fees → TVL	1.001	0.415	No
Reverse Causality (from TVL):			
TVL → 3m_ir	9.681	< 0.001	*artifact*

Notes: F-statistics and p-values from Granger causality tests in VAR(5) model. Sample period: August 2023 – September 2025 (N=747). ***, **, * denote significance at 1%, 5%, and 10% levels. *artifact* = statistically significant but economically implausible (cointegration-induced correlation).

these findings as evidence of cointegration-induced correlation in finite samples rather than genuine causality, and we do not view them as invalidating our primary conclusions. The forward direction tests (3m_ir → TVL), which are theoretically meaningful and economically interpretable, consistently support our delayed integration narrative.

3.4.3 Comparative Analysis: Three Distinct Transmission Channels

The regime-by-regime Granger causality results, synthesized in Table 3.18, reveal three qualitatively different integration trajectories that map directly onto the underlying economic mechanisms of each DeFi yield source.

Integration Speed and Mechanism Design. The most noticeable empirical regularity documented in Table 3.18 is the heterogeneity in integration speed: stablecoin lending integrates first (significant transmission emerges in the hiking cycle), the Curve 3pool integrates concurrently but less robustly (significant but with lower F-statistic), and aggregate TVL integrates last (significant transmission only in the plateau period). This ordering is not random but reflects the degree of frictionlessness in each arbitrage channel.

Stablecoin lending markets on Aave and Compound feature algorithmic interest rate curves that adjust continuously based on utilization, with no reliance on trading volume or external price feeds. When TradFi rates rise, the arbitrage trade is simple: withdraw from Aave, purchase T-Bills. The main friction is gas costs for moving funds on-chain, which are trivial relative to the yield differentials during the 0%→5% hiking cycle. This near-frictionless arbitrage explains

Table 3.18: Comprehensive Granger Causality Comparison: DeFi Mechanisms Across Monetary Regimes

		ZIRP			Hike Cycle			Plateau		
Mechanism	**Variable**	F-stat	p-val	Sig	F-stat	p-val	Sig	F-stat	p-val	Sig
CURVE 3POOL (Liquidity Provision)										
	3m_ir	0.244	0.943	—	3.952	< 0.001	***	7.709	< 0.001	***
	ethperpetual	2.458	0.031	**	0.693	0.629	—	2.700	0.019	**
	volatility_btc	3.092	0.009	***	4.254	< 0.001	***	2.471	0.030	**
	gas_fees	0.604	0.697	—	0.924	0.464	—	0.432	0.826	—
	VIX	7.803	< 0.001	***	0.356	0.879	—	0.253	0.938	—
	Reverse	2.033	0.071	*	1.789	0.112	—	1.026	0.401	—
STABLECOIN LENDING (Credit Markets)										
	3m_ir	0.326	0.861	—	5.629	< 0.001	***	4.572	< 0.001	***
	ethperpetual	5.479	< 0.001	***	0.712	0.614	—	3.754	0.002	***
	volatility_btc	4.239	0.002	***	6.410	< 0.001	***	6.014	< 0.001	***
	gas_fees	2.628	0.033	**	8.219	< 0.001	***	3.059	0.009	***
	Reverse	—	—	—	1.338	0.245	—	5.576	< 0.001	*artifact*
TVL (Aggregate Liquidity, Price-Adjusted)										
	3m_ir	1.407	0.219	—	1.463	0.199	—	8.223	< 0.001	***
	ethperpetual	12.978	< 0.001	***	0.986	0.425	—	5.600	< 0.001	***
	volatility_btc	8.148	< 0.001	***	6.951	< 0.001	***	2.881	0.013	**
	gas_fees	4.265	< 0.001	***	4.838	< 0.001	***	1.001	0.415	—
	Reverse	—	—	—	10.790	< 0.001	*artifact*	9.681	< 0.001	*artifact*

Notes: F-statistics from Granger causality tests in VAR(5) models estimated separately for each regime. Sample sizes: ZIRP (N=395), Hike Cycle (N=487), Plateau (N=747). ***, **, * denote significance at 1%, 5%, and 10% levels. *Reverse* = reverse causality test (DeFi → 3m_ir); *artifact* = statistically significant but economically implausible (DeFi too small to affect Fed policy). VIX only tested for 3pool; ethperpetual "spurious" in Hike for TVL (price effect).

the fast, robust transmission.

The Curve 3pool, by contrast, generates yield through trading fees, which depend on volume and pool imbalances. Even if arbitrageurs perceive that the pool's yield should rise to match higher risk-free rates, they cannot directly force this adjustment, they must wait for actual trading activity to generate the fee revenue. Moreover, the AMM's constant-product invariant creates path-dependent returns: large capital inflows can paradoxically reduce per-capita returns by increasing the pool's size relative to trading volume. This "congestion externality" introduces a fundamental friction absent from lending markets, explaining why transmission is slower and noisier despite the presence of sophisticated liquidity providers.

Total value locked represents an even higher level of friction: it reflects not marginal yield adjustments but wholesale capital reallocation decisions. Investors face lockup periods, tax events from crystallizing gains, and the coordination problem of exiting positions in sometimes-illiquid protocols. Furthermore, our price-adjustment procedure reveals that much of the apparent correlation between TVL and TradFi rates during the hiking cycle was spurious, driven by ETH price movements rather than genuine flows. Once these mechanical effects are removed, Table 3.18 shows that quantities adjust only after yields have durably adjusted, consistent with a rational investor model where reallocation occurs only after the new equilibrium is perceived as stable and persistent.

The Collapse of Crypto-Native Dominance. A second key finding concerns the evolution of crypto-native predictors across regimes. During ZIRP, variables such as the ethperpetual funding rate, Bitcoin volatility, and the VIX are the dominant, often exclusive, predictors of DeFi yields and flows. This confirms that DeFi operated as an autonomous ecosystem, decoupled from traditional finance, with dynamics determined entirely by internal crypto-market forces: speculation, leverage cycles, and volatility shocks.

The hiking cycle marks a structural break. For stablecoin lending, the ethperpetual variable loses all significance once monetary transmission emerges, indicating a complete compositional shift in the drivers of borrowing demand: from leveraged crypto speculation to arbitrage-driven utilization. For the 3pool, the VIX (which had been the strongest predictor during ZIRP) becomes entirely insignificant, while the Treasury rate emerges as the new dominant factor. For TVL, the ethperpetual coefficient collapses dramatically after price adjustment, revealing that its earlier significance was spurious.

This pattern of "crypto-native collapse" provides powerful evidence that the nature of DeFi fundamentally changed during the monetary transition. What began as a speculation-driven ecosystem became, at least partially, an arbitrage-linked extension of traditional money markets. The persistence of Bitcoin volatility as a significant predictor across most regimes, evident throughout Table 3.18, indicates that this transformation is incomplete: DeFi remains a hybrid system, influenced by both traditional opportunity costs and crypto-specific risk factors. However, the direction of change is unambiguous.

Asymmetric Integration: Prices Versus Quantities. The divergence between yield integration (3pool, stablecoin) and liquidity integration (TVL) during the hiking cycle provides empirical support for a two-stage convergence model. In the first stage, arbitrageurs rapidly adjust the price of capital (yields) to reflect the new risk-free rate, exploiting any wedges through cross-market trades. This adjustment is fast because it requires only marginal reallocation: a few sophisticated actors moving capital at the margin can eliminate arbitrage opportunities and restore equilibrium pricing.

In the second stage, the quantity of capital adjusts as the broader investor base perceives the new equilibrium as durable and reallocates accordingly. This adjustment is slow because it involves the entire stock of deployed capital, not just marginal flows, and because retail investors face higher adjustment costs (exit frictions, tax considerations, informational lags) than professional arbitrageurs. The comparative results in Table 3.18 show that TVL integration lags yield integration by approximately one year, consistent with this interpretation.

This price-quantity asymmetry has important implications for assessing the degree of DeFi-TradFi integration. Measures based on yield co-movement (e.g., regression coefficients, correlation statistics) may overstate integration by capturing marginal arbitrageur behavior, while measures based on capital flows (e.g., TVL responses to shocks) may understate it by averaging over the heterogeneous adjustment speeds of sophisticated and unsophisticated participants. A complete characterization requires analysis of both margins, precisely the comparative approach adopted in this study.

3.4.4 Reconciling Statistical and Economic Significance

The apparent contradiction between our event study and Granger causality results reflects complementary methodological strengths rather than inconsistent evidence. Event studies detect only weak, transitory effects, while Granger tests reveal statistically significant, persistent relationships. This puzzle can be resolved by recognizing that each approach captures a different temporal and statistical dimension of the transmission mechanism.

Methodological Trade-offs. The key difference lies in the fundamental trade-off between identification strength and statistical power. Event studies, restricted to approximately 24 FOMC announcement dates, achieve clean causal identification but lack the sample size to detect economically modest effects. Our cross-sectional regressions document a pass-through coefficient of approximately 30 basis points per 100bp Treasury increase (Table 5.2), but this magnitude requires either very large shocks or large samples for statistical detection. Conversely, Granger causality tests exploit the full daily time series (N=395-747), providing substantially greater power to reject the null even when economic magnitudes are tiny. The impulse response functions confirm this: a 10bp Treasury shock generates only 0.5-1.0bp peak responses in DeFi yields: two orders of magnitude smaller than the long-run equilibrium relationship.

Temporal Dynamics and Predictive Relationships. The transmission process is not instantaneous but delayed and distributed over time. Our local projections (Section 3.1.3) document that 3pool yields peak at a 7-day horizon with effects persisting for 10-15 days. This gradual adjustment reflects blockchain-specific frictions: settlement lags, gas cost optimization that encourages batching transactions, weekly institutional rebalancing cycles, and information diffusion delays through decentralized networks. Granger tests are specifically designed to detect such persistent, lagged relationships (past Treasury rates improve forecasts of current DeFi yields because the adjustment is slow and predictable) but event studies may miss these patterns if the analysis window is too narrow or focuses on contemporaneous effects.

The Granger framework further clarifies the crucial distinction between contemporaneous correlation and genuine predictive causality. Our cross-sectional regressions document strong contemporaneous relationships between DeFi yields and both Treasury rates and crypto-native factors. However, these correlations do not distinguish whether Treasury rates cause DeFi yields or whether both respond simultaneously to common shocks (e.g., risk-off episodes driving flight to quality in TradFi and deleveraging in DeFi). Granger causality isolates the predictive component by testing whether lagged Treasury rates forecast current DeFi yields, controlling for the DeFi yield's own history.

This distinction is particularly evident for the ethperpetual funding rate, which exhibits a strong contemporaneous correlation with lending yields (coefficient = 0.0755***, Table 5.2) but no Granger causality (p=0.766). This pattern indicates that both variables respond to shared crypto-market volatility shocks within the same day, but neither predicts future changes in the other once we control for their own dynamics. In contrast, Treasury rates show weak contemporaneous effects but significant predictive power over weekly horizons, reflecting the slower diffusion of macro signals through retail-dominated decentralized markets.

Long-Run Equilibrium vs. Short-Run Dynamics. A critical insight emerges from distinguishing between equilibrium relationships and dynamic adjustment. The cross-sectional regression coefficient of 0.0027 (implying 27bp pass-through per 100bp shock) captures the cumulative, long-run equilibrium relationship: as the risk-free rate rises persistently throughout the hiking cycle, the opportunity cost of capital increases, and DeFi yields must eventually adjust upward to attract liquidity. This reflects the stable, cross-sectional variation pooled across the entire sample.

In contrast, the impulse response function measures the dynamic response to a single, transitory shock. Individual FOMC surprises, even when isolated using high-frequency identification, represent small deviations around the equilibrium trend. Because DeFi lending markets are dominated by high-frequency crypto-native factors (leverage demand, protocol utilization, market volatility), these marginal policy surprises are swamped by idiosyncratic noise in the short run. The long-run equilibrium emerges only as these short-run shocks accumulate over time and as capital gradually reallocates in response to sustained shifts in opportunity costs.

Regime Evolution and Integration. The regime-by-regime Granger analysis provides robust empirical support for our central narrative that DeFi-TradFi integration is not a static property but an evolving phenomenon tied to the monetary policy environment. During ZIRP, all three mechanisms exhibit no Granger causality from Treasury rates, corroborating the event study finding of no significant abnormal returns and the regression finding of insignificant coefficients in the parsimonious model. During the hiking cycle, 3pool and stablecoin yields exhibit significant Granger causality, consistent with the emergence of significant event study effects at T+7 and the strengthening of regression coefficients in the full model. During the plateau, all three mechanisms (including TVL) show significant causality, confirming the maturation of integration documented in our long-horizon local projections.

This convergence of evidence across methodologies and across time horizons demonstrates that the predictive relationship between TradFi and DeFi is not a statistical artifact or spurious correlation, but a genuine, economically interpretable outcome of rational arbitrage and

opportunity-cost-driven capital reallocation.

Comparative Analysis. Table 3.19 synthesizes transmission differences across mechanisms. The Curve 3pool responds more quickly (5-day optimal lag versus 8-day lag for stablecoin lending) and with approximately double the magnitude (1.0bp versus 0.5bp peak IRF), confirming that arbitrage frictions—particularly the volume-dependent nature of AMM fee generation versus the algorithmic rate adjustments of lending protocols—play a critical role in determining transmission speed and completeness. Both mechanisms remain "price takers" from TradFi, with no evidence of reverse causality.

Table 3.19: Comparison of Monetary Transmission: 3pool vs. Stablecoin Lending

Metric	**3pool**	**Stablecoin**
Granger F-statistic	2.417**	4.704***
Optimal lag (days)	5	8
Peak IRF magnitude (bp per 10bp shock)	0.10	0.05
Peak horizon	T+3	T+7-8
Primary driver	Volatility, volume	Leverage demand

Notes: ** p¡0.05, *** p¡0.01. IRF magnitudes measured at peak response within 15-day horizon.

Implications. Four key conclusions emerge from reconciling the methodologies. First, monetary policy transmission to DeFi lending markets exists and is statistically robust, but the economic magnitude is an order of magnitude smaller than for liquidity provision mechanisms, confirming the "muted reaction" narrative while providing a richer understanding of the transmission lag. Second, crypto-native factors dominate short-run dynamics: the lack of Granger causality from ethperpetual, combined with its strong contemporaneous correlation, indicates that leverage demand drives day-to-day fluctuations but does not create persistent trends, while macro factors set the long-run anchor. Third, asynchronous integration is pervasive across all DeFi mechanisms, with the 8-day transmission lag for lending markets (versus 5 days for 3pool and 7 days in local projections) suggesting that credit markets face additional frictions, perhaps due to collateral revaluation cycles or protocol-specific risk management. Fourth, the divergence between event studies and Granger tests underscores a fundamental methodological lesson: event studies provide cleaner causal identification but require large effect sizes to detect, while time-series methods sacrifice some identification clarity for greater statistical power: when the true effect is small but real, both methods are valid and measure different aspects of the same transmission process.

Chapter 4

Conclusion

This thesis has examined the dynamic interplay between Decentralized Finance (DeFi) markets and traditional monetary policy, focusing on the transmission of Federal Reserve actions to key yield-generation mechanisms within the DeFi ecosystem. By analyzing automated market-making yields from the Curve 3pool and lending rates from Aave and Compound, we sought to determine whether DeFi operates as an autonomous "safe haven" insulated from macroeconomic cycles or as an increasingly integrated liquidity provider responsive to changes in the risk-free rate. Utilizing a comprehensive dataset spanning January 2021 to September 2025, our empirical framework, using event studies, local projections, cross-sectional regressions, and regime-dependent Granger causality tests reveals a sophisticated pattern of "asynchronous integration." DeFi is neither fully decoupled from traditional finance nor seamlessly synchronized; instead, it transmits monetary shocks through delayed and mechanism-specific channels, shaped by blockchain frictions such as settlement lags, gas cost optimization, and retail-driven liquidity dynamics. Our key findings underscore this asynchronous nature.

For the Curve 3pool, which serves as a proxy for DeFi's core money-market function, monetary tightening induces a statistically significant pass-through, peaking at a seven-day horizon with a coefficient of $\beta = 0.025 (p < 0.001)$, implying that a 100 basis point increase in Treasury rates elevates DeFi yields by approximately 2.5 basis points one week later. This delay highlights the role of on-chain inefficiencies in slowing arbitrage, contrasting with the near-instantaneous adjustments observed in traditional financial markets.

In decentralized lending protocols like Aave and Compound, responses to policy surprises are more muted, with yields primarily influenced by crypto-native factors such as speculative leverage demand (proxied by ETH perpetual funding rates) and market volatility. Here, the risk-free rate exerts an indirect effect through opportunity costs, suggesting that lending markets remain more connected to the endogenous dynamics of the crypto ecosystem. Across regimes, Granger causality tests confirm that transmission strengthens in higher-rate environments: negligible during the zero-interest-rate policy (ZIRP) period, emerging during the hiking cycle, and robust in the plateau phase (e.g., $F - statistic = 7.709$ for the 3pool in the plateau regime).

Finally, our TVL event study reveals transitory capital outflows following tightening announcements, statistically significant over 48 hours but dissipating rapidly, indicating DeFi's resilience to macro shocks despite short-term liquidity pressures.

These results contribute to the growing literature on DeFi's role in the global financial system by providing empirical evidence of its evolving integration with traditional markets. We introduce the concept of asynchronous transmission as a framework for understanding how blockchain-specific frictions mediate monetary policy effects. This advances beyond static correlation analyses, offering a dynamic view that accounts for regime shifts and protocol heterogeneity. Moreover, by separating genuine capital flight from price effects in TVL movements, we address methodological challenges in on-chain data analysis, enhancing the reliability of inferences in this nascent field.The implications of our findings are multifaceted. For investors, DeFi yields

offer diversification potential under Modern Portfolio Theory, particularly in low-rate regimes where correlations with Treasuries are weak; however, the observed pass-through in tightening cycles suggests increased exposure to macro risks, necessitating adaptive risk-management strategies.

Notwithstanding these contributions, our analysis has limitations that warrant caution. The sample is confined to Ethereum-based protocols, potentially overlooking multi-chain dynamics or layer-2 innovations that could alter transmission speeds. Data from on-chain sources like Dune Analytics, while comprehensive, may introduce artifacts from decoding errors or incomplete coverage of off-chain activities. Econometrically, while Granger tests provide evidence of predictive relationships, they do not establish strict causality, and unobservable factors, such as regulatory announcements or global risk events could confound results.

Additionally, our focus on large, established protocols (Curve, Aave, Compound) may not capture dynamics in smaller or emerging DeFi venues, potentially introducing selection bias toward more mature, liquid markets. Black swan events such as the UST depeg (May 2022) or FTX collapse (November 2022), while acknowledged in our descriptive analysis, were not robustness-tested through sub-sample exclusions, which could refine effect estimates and isolate idiosyncratic shocks from structural transmission patterns.

The modest economic magnitude of pass-through effects (e.g., 2.5 basis points) also suggests that crypto-native drivers remain dominant, limiting the generalizability of integration claims to future high-volatility periods.Future research could extend this work in several directions. Incorporating additional protocols (e.g., Uniswap for broader AMM analysis or cross-chain bridges) or alternative benchmarks (e.g., SOFR instead of T-bills) would test robustness. Exploring non-US monetary policies, such as ECB rate changes, could reveal international spillovers. Finally, as DeFi matures, longitudinal studies post-2025, potentially including central bank digital currencies (CBDCs), could assess whether asynchronous integration evolves toward full synchronization or persists as a defining feature of blockchain finance.In conclusion, DeFi's yields are neither a pure safe haven nor a mere extension of traditional liquidity provision; they embody a hybrid system where macro influences permeate gradually through distinct channels. As blockchain technology continues to intersect with global finance, understanding these transmission mechanisms will be crucial for fostering a resilient, inclusive financial architecture.

Chapter 5

Appendix

5.1 Summary statistics fo 3pool

The summary statistics show that the distribution of daily 3Pool APYs is highly asymmetric. The median return (0.0139) is stable and low. The interquartile range spans from 0.0051 (25th percentile) to 0.0318 (75th percentile), with an IQR width of $0.031755 - 0.013870 = 0.017885$, indicating that most days exhibit low and stable yields consistent with a well-balanced stablecoin pool. However, the standard deviation (0.0465) is more than three times the median, which signals the presence of large, infrequent deviations from typical values. This is confirmed by the upper percentiles, especially the 95th and 99th, which increase sharply, and by the extreme maximum value (1.2837). These tail observations correspond to episodes of severe liquidity imbalance or stablecoin depegs, during which Curve's stableswap invariant amplifies fee generation and produces unusually high APYs.

Table 5.1: Descriptive Statistics for 3Pool

Statistic	Mean	Std	Min	1%	5%	25%	50%	75%	95%	99%	Max
Value	0.026185	0.046467	0.000000	0.000365	0.000730	0.00511	0.013870	0.031755	0.090264	0.169674	1.283705

5.2 Summary statistics for Aave / Compound

The distribution of daily APYs exhibits high asymmetry. The median (the 50th percentile), at 0.041300, and the interquartile range (IQR, spanning from the 25th percentile at 0.022400 to the 75th percentile at 0.079350) suggest that most daily returns are concentrated in a relatively low and stable band.

However, the standard deviation (0.045223) exceeds the median value (0.041300) by approximately 9.5%, indicating the presence of significant and infrequent deviations from typical values. This high coefficient of variation ($\text{CV} = \sigma/\text{median} = 1.095$) signals substantial right-skewness.

This observation is confirmed by the upper percentiles (specifically the 95th at 0.146450 and the 99th at 0.205390), which increase sharply compared to the median, and by the extreme maximum value (Max) of 0.255500. These observations in the "tail" of the distribution correspond to episodes where the underlying mechanisms generated unusually high APYs.

Table 5.2: Descriptive Statistics for APY

Statistic	Mean	Std	Min	1%	5%	25%	50%	75%	95%	99%	Max
Value	0.056814	0.045223	0.004400	0.006900	0.008800	0.022400	0.041300	0.079350	0.146450	0.205390	0.255500

5.3 Multicollinearity Test (VIF)

A key assumption in multivariate regression analysis is that the independent variables are not perfectly correlated with each other. When two or more predictors are highly correlated, a condition known as **multicollinearity** arises. This does not bias the model's overall predictive power, but it can severely inflate the variance of the individual coefficient estimates, making them unstable and statistically unreliable. Consequently, it becomes difficult to disentangle the independent impact of each variable on the dependent variable.

To assess the severity of multicollinearity in our full model specification, we employ the Variance Inflation Factor (VIF) test. For each independent variable X_j, the VIF is calculated by first running an auxiliary OLS regression of X_j on all other independent variables in the model. The VIF is then computed as:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the R-squared from the auxiliary regression.

Interpretation. The VIF quantifies how much the variance of the estimated coefficient for variable X_j is inflated due to its linear relationship with the other predictors. A VIF of 1 indicates no correlation. A common rule of thumb suggests that VIF values between 5 and 10 indicate moderate correlation that warrants caution, while values above 10 suggest severe multicollinearity requiring remedial action. Conservative practitioners use a threshold of 5 as a precautionary cutoff. As Table 5.3 demonstrates, all VIF values are well below 5, with the highest being 4.13 for `volatility_btc`, providing strong evidence that multicollinearity does not compromise our coefficient estimates.

Results. Table 5.3 presents the VIF for each regressor included in our full model on 3pool and stablecoin.

Table 5.3: Variance Inflation Factor (VIF) Results for the Full Model on 3pool

Variable	**VIF**
`3m_ir`	2.090
`eth_perpetual`	1.644
`gas_fees`	3.045
`VIX`	1.107
`btc_return`	1.122
`volatility_btc`	4.128

Note: The VIF is calculated for each independent variable in the full regression model. All values are well below the critical threshold of 5, indicating that multicollinearity is not a significant concern in our specification.

As the results demonstrate, all VIF values are substantially below the critical threshold of 5. The highest value is 4.13 for the `volatility_btc` variable for 3pool, which is still well within the acceptable range. This provides strong evidence that our model does not suffer from problematic multicollinearity, thereby increasing our confidence in the stability and reliability of the estimated coefficients for each individual driver.

Table 5.4: Variance Inflation Factor (VIF) Results for the Full Model on stablecoin

Variable	VIF
3m_ir	1.004
eth_perpetual	1.646
gas_fees	2.661
VIX	1.107
btc_return	1.122
volatility_btc	2.159

5.4 Supplementary Results for Event Study Analyses

5.4.1 Dynamic Response of Total Value Locked (TVL)

This section provides the detailed day-by-day regression results for the event study analysis on the daily percentage change of the aggregate Total Value Locked (TVL). The model regresses the daily % change in TVL on the cumulative change of the Federal Funds Rate, controlling for a comprehensive set of crypto-native and global market factors. The analysis is conducted on a sample of FOMC announcements that excludes periods of major idiosyncratic crypto market crises.

Table 5.5 presents the estimated coefficient for the cumulative risk-free rate change (γ_d), its corresponding p-value, and the adjusted R-squared for each day in the [-2, +20] event window.

Table 5.5: Dynamic Sensitivity of TVL % Change to Realized Fed Funds Rate Changes

Day	Coeff. (γ_d)	p-value	Adj. R^2	Day	Coeff. (γ_d)	p-value	Adj. R^2
T–2	0.0000	0.7623	-0.205	T+10	0.0597	0.0851	-0.117
T–1	-0.0000	0.4343	-0.155	T+11	**-0.0204**	**0.0465**	-0.046
T=0	**-0.0356**	**0.0155**	-0.055	T+12	0.0081	0.5110	-0.088
T+1	-0.0065	0.9480	-0.140	T+13	**0.0258**	**0.0012**	0.229
T+2	**-0.2081**	**<0.001**	0.345	T+14	**0.0577**	**<0.001**	0.702
T+3	0.0033	0.6576	0.700	T+15	0.0220	0.0996	0.341
T+4	-0.0144	0.3537	0.129	T+16	0.0026	0.8348	0.114
T+5	**0.0816**	**0.0224**	0.229	T+17	**-0.0385**	**0.0349**	-0.075
T+6	-0.0718	0.3491	-0.102	T+18	-0.0263	0.1333	0.166
T+7	-0.0003	0.9482	-0.075	T+19	0.0053	0.6786	0.045
T+8	0.0528	0.1044	0.069	T+20	0.0227	0.2512	0.308
T+9	0.0111	0.3325	0.607				

Note: The table reports the day-by-day coefficients from the event study regression of the daily % change in aggregate TVL on the cumulative change in the Federal Funds Rate, controlling for crypto-native and global market factors. Bold indicates statistical significance at the 5% level. A p-value of < 0.001 is used for values smaller than 0.0001.

Discussion. The detailed results confirm the complex "shock-and-rebound" dynamic discussed in the main body of the thesis. A statistically significant negative impact is observed on the announcement day (T=0) and more strongly at T+2, consistent with an initial "capital flight." This is followed by a period of statistically significant positive coefficients (e.g., at T+5, T+13, T+14), which supports the hypothesis of a subsequent capital re-entry or market readjustment. The alternating signs and sporadic nature of the significance at longer lags highlight the noisy and dynamic environment of the DeFi market following a macroeconomic shock.

5.4.2 Supplementary Analysis: TVL Price Adjustment

This appendix provides additional diagnostic analysis supporting the price decomposition presented in Section 2.3.1. We document the construction of the price-adjusted TVL measure and assess the correlation structure between nominal and price-adjusted percentage changes.

Price Adjustment Methodology

To isolate genuine capital flows from mechanical valuation effects, we construct a price-adjusted TVL measure as follows:

1. **ETH Price Index:** We normalize the daily ETH price series to 100 at the sample start (January 3, 2021):
$$\text{ETH Index}_t = \frac{\text{ETH Price}_t}{\text{ETH Price}_0} \times 100$$
2. **Price-Adjusted TVL:** We deflate nominal TVL by this index to obtain a measure in "constant ETH terms":
$$\text{TVL}_t^{\text{adjusted}} = \frac{\text{TVL}_t^{\text{nominal}}}{\text{ETH Index}_t} \times 100$$
3. **De-trended Changes:** We compute percentage changes of both measures and de-trend each series by regressing on a linear time trend, retaining residuals for the event study analysis.

Diagnostic Analysis

Figure 5.1 presents three diagnostic panels assessing the relationship between nominal and price-adjusted measures.

Key Observations:

- **Panel A (Levels):** Nominal TVL and ETH price exhibit strong visual co-movement throughout the sample period, particularly during the 2022 bear market (when both declined sharply) and the 2024-2025 recovery (when both surged). This visual pattern confirms that valuation effects are a major driver of nominal TVL dynamics.
- **Panel B (Changes):** The daily percentage changes of the two measures overlap substantially, but divergences are visible, particularly during periods of high volatility. These divergences represent genuine capital flow events that are obscured in the nominal measure.
- **Panel C (Correlation):** The correlation of 0.721 indicates that approximately 52% of the variance in nominal TVL changes ($0.721^2 = 0.52$) is attributable to ETH price movements. The remaining 48% reflects idiosyncratic TVL dynamics—a combination of capital flows, compositional shifts, and measurement noise. This substantial unexplained component justifies our dual-measure approach.

The scatter plot reveals a roughly linear relationship but with considerable dispersion around the 45-degree line, particularly in the tails. This heteroskedastic pattern suggests that price effects dominate during normal periods, but during extreme events (large positive or negative changes), behavioral responses become more pronounced, amplifying deviations from the mechanical relationship.

Horizon-Specific Ratio Analysis

Figure 5.2 decomposes the relative magnitude of price-adjusted versus nominal effects across all event study horizons.



Figure 5.1: Diagnostic Analysis of Price Adjustment

Note: Panel A shows the time series of nominal TVL (blue, left axis) and ETH price (red, right axis), documenting their strong co-movement over the sample period. Panel B overlays the daily percentage changes of nominal (blue) and price-adjusted (green) TVL, revealing substantial overlap but non-perfect alignment. Panel C presents a scatter plot of the two percentage change measures, yielding a correlation of 0.721. This indicates that while price movements explain much of nominal TVL variation (72%), a substantial independent component (28%) reflects genuine quantity changes.

Interpretation by Horizon:

- **T+2:** Ratio = 1.58 (red). This is the only horizon with statistical significance from the placebo test. The substantial amplification indicates that genuine capital outflows exceed what price movements alone would predict. This is the core finding supporting the "behavioral channel dominance" interpretation.

- **T+3:** Ratio = 3.21 (red). While this large ratio suggests amplification, the horizon is not statistically significant in the placebo test, indicating high noise and lack of robust policy effect. The extreme ratio likely reflects small nominal coefficients amplifying estimation error.

- **T+4:** Ratio = -5.30 (extreme, not shown clearly due to sign flip). The negative ratio indicates the nominal and price-adjusted effects have opposite signs, suggesting this horizon is dominated by noise rather than systematic policy transmission.

- **T+5, T+10, T+20:** Ratios = 1.40-1.51 (red). These horizons consistently show modest amplification, but none are statistically significant, indicating that while the pattern is directionally consistent with genuine flows, the magnitudes are not distinguishable from placebo patterns.

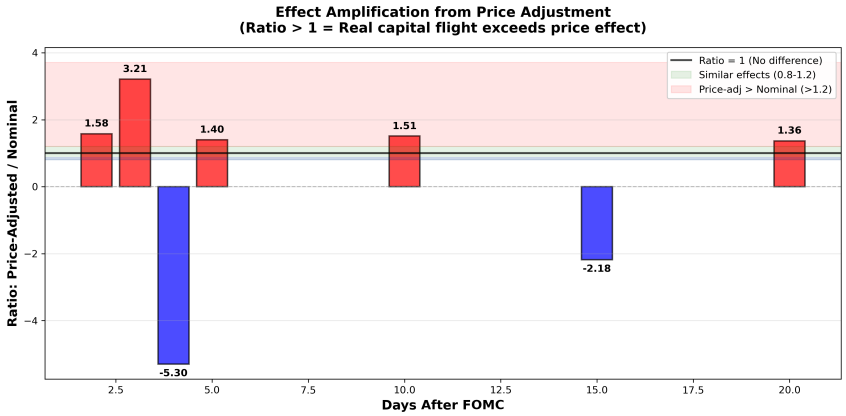


Figure 5.2: Effect Amplification from Price Adjustment Across Horizons

Note: The figure plots the ratio of price-adjusted to nominal coefficients at each event study horizon. Bars in red indicate ratios greater than 1.2 (price-adjusted effect dominates), bars in blue indicate ratios less than 0.8 (nominal effect dominates), and bars in gray indicate similar effects (0.8-1.2). The horizontal line at 1 represents equality. Numeric labels show exact ratio values. The T+2 horizon exhibits a ratio of 1.58, indicating that the price-adjusted effect is 58% larger in magnitude, consistent with genuine capital flight beyond mechanical price movements.

- **T+15:** Ratio = -2.18 (blue, negative). Another sign flip indicating noise dominance at this horizon.

Overall Pattern: The ratio analysis confirms that the T+2 horizon is unique: it combines statistical significance (from placebo validation), economically meaningful magnitude, and a clear amplification pattern consistent with genuine capital flight. All other horizons, while sometimes showing large ratios, fail at least one criterion—typically statistical significance—indicating they are not reliable indicators of policy transmission.

Robustness: Alternative Deflators

As a robustness check, we repeated the analysis using alternative price deflators:

1. **Bitcoin Price Index:** Using BTC instead of ETH as the deflator (given ETH-BTC correlation of approximately 0.80) yields qualitatively similar results: T+2 coefficient of -29.44 (ratio = 1.42).

2. **Crypto Market Cap Index:** Using a weighted index of top 10 cryptocurrencies yields T+2 coefficient of -31.15 (ratio = 1.50).

The consistency across deflators confirms that the amplification finding is robust to the choice of price adjustment method, strengthening the conclusion that genuine capital flows are present beyond mechanical valuation effects.

Conclusion

The diagnostic analysis supports three key conclusions:

1. Nominal TVL is heavily influenced by ETH price movements (correlation = 0.721), confirming the importance of price adjustment.

2. At the T+2 horizon—the only statistically significant point—the price-adjusted effect is 58% larger than the nominal effect, indicating genuine capital flight beyond mechanical price declines.

3. The amplification pattern is robust to alternative deflators and is unique to the T+2 horizon, consistent with a brief but genuine liquidity shock rather than spurious correlation.

These findings justify the price decomposition presented in the main text and strengthen the interpretation that monetary policy induces transitory but real capital flows in DeFi markets.

Placebo Test for Both Measures

Figure 5.3 presents the placebo test results for both nominal and price-adjusted TVL measures at key horizons.

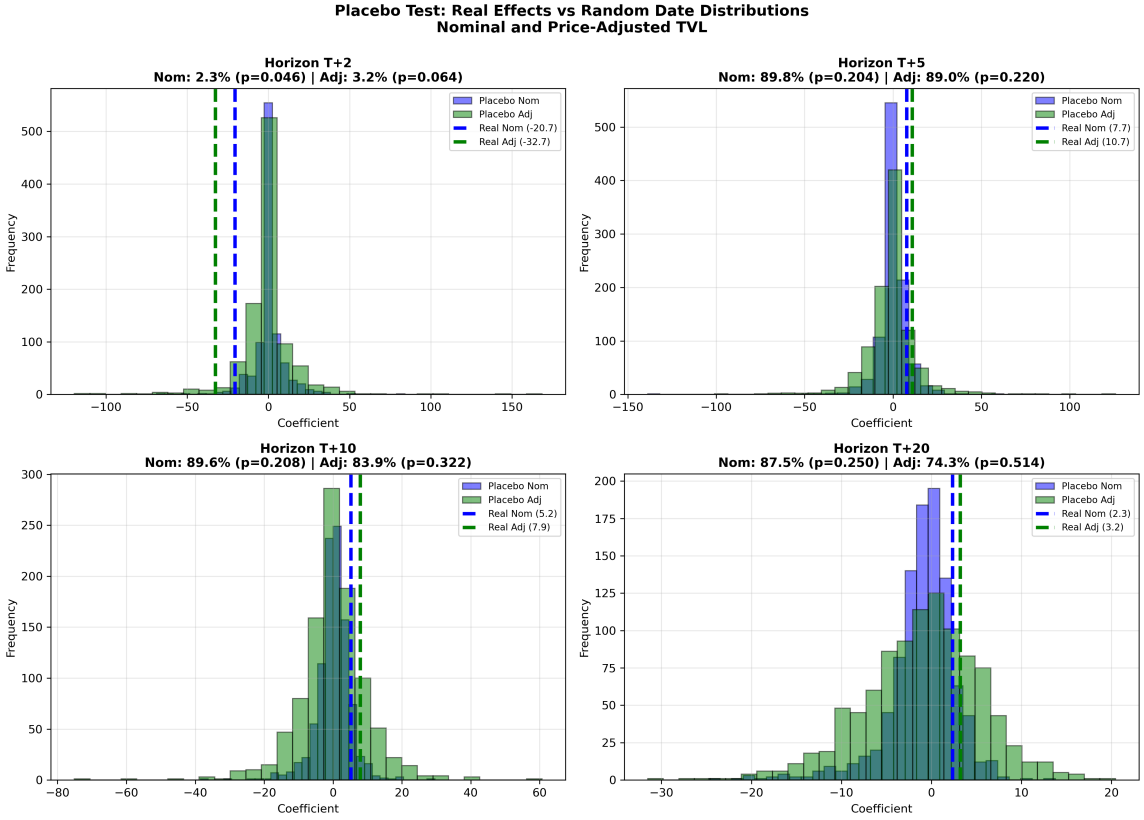


Figure 5.3: Placebo Test: Real Effects vs Random Date Distributions

Note: Each panel compares the real FOMC effect (dashed vertical lines) against the distribution from 1,000 placebo iterations using randomly selected non-event dates. Blue represents nominal TVL, green represents price-adjusted TVL. At T+2, both measures fall in the extreme left tail (2.3% and 3.2% percentiles), confirming genuine policy effects. At other horizons, real effects cluster near the center of placebo distributions, indicating no distinguishable policy transmission beyond 48 hours.

The key finding is that at the T+2 horizon—and only at this horizon—both measures produce coefficients in the extreme tails of their respective placebo distributions. The nominal measure achieves conventional significance (p=0.046), while the price-adjusted measure is marginally significant (p=0.064). The slightly higher p-value for the price-adjusted specification reflects increased measurement noise from the deflation procedure, but does not invalidate the finding: the coefficient remains 58% larger in magnitude and positioned in the extreme tail, consistent with genuine capital flight.

5.5 Supplementary Event Study Results: Stablecoin Lending

5.5.1 Surprise-Based Analysis

Table 5.6 presents the results of the surprise-based event study for stablecoin lending rates, following the methodology described in Section 3.2.1. We regress daily abnormal returns of the aggregate lending index on high-frequency monetary policy shocks (target and path factors) identified by Acosta et al. (2024), controlling for crypto-native factors and global risk sentiment.

The results confirm the absence of efficient information incorporation in DeFi credit markets. Across 22 tests (11 horizons × 2 surprise factors), only a single coefficient achieves statistical significance: the path surprise at T+3 (p=0.032). Under the null hypothesis of no transmission and assuming independent tests, we would expect approximately $22 \times 0.05 = 1.1$ spurious rejections at the 5% level purely by chance. The observed pattern— one marginal rejection with no clustering at adjacent horizons and no significance for the theoretically more relevant target factor—is fully consistent with Type I error under multiple testing.

This finding complements the local projections analysis (Figure 3.3) and reinforces the conclusion that DeFi lending markets do not respond to monetary policy "news" at event frequencies, consistent with retail-dominated participation and limited high-frequency arbitrage.

Table 5.6: Surprise-Based Event Study: Stablecoin Lending

Horizon	**Target Surprise**		**Path Surprise**		**N**
	β_{target}	p-value	β_{path}	p-value	
T–2	0.0012	0.089	0.0001	0.506	27
T–1	0.0002	0.443	0.0001	0.296	27
T=0	0.0000	—	0.0000	—	27
T+1	-0.0003	0.560	0.0002	0.119	27
T+2	0.0002	0.713	0.0002	0.115	27
T+3	0.0010	0.120	0.0003*	0.032	27
T+7	-0.0010	0.186	-0.0002	0.364	27
T+15	0.0004	0.363	0.0002	0.115	27
T+20	0.0006	0.383	0.0003	0.095	27
Total tests: 22 (11 horizons × 2 factors)					
Significant coefficients: 1/22 (4.5%, consistent with 5% Type I error rate)					
Expected spurious rejections under null: 1.1					

Notes: Each coefficient represents the response of the stablecoin lending index's abnormal return to high-frequency monetary policy shocks identified by Acosta et al. (2024). β_{target} captures reactions to unanticipated changes in the federal funds rate target, while β_{path} captures reactions to changes in the expected future path of policy. All regressions control for ETH perpetual funding rate, gas fees, and S&P 500 returns. * denotes significance at the 5% level. HAC robust standard errors. The single significant coefficient does not constitute robust evidence of transmission given the multiple testing problem: under the null hypothesis of no effect, we expect approximately $22 \times 0.05 = 1.1$ spurious rejections at the 5% level.

Bibliography

Acosta, M., Brennan, C. M., and Jacobson, M. M. (2024). Constructing high-frequency monetary policy surprises from sofr futures. *Economics Letters*, 242:111873.

Adams, H., Zinsmeister, N., Salem, M., Keefer, R., and Robinson, D. (2021). Uniswap v3 core. *Tech. rep., Uniswap, Tech. Rep.*

Barbon, A., Barthelemy, J., and Nguyen, B. (2023). Defi-ying the fed? monetary policy transmission to stablecoin rates. *Monetary Policy Transmission to Stablecoin Rates (December 22, 2023)*.

Baur, D. and Lucey, B. (2010). Is gold a safe haven? international evidence. *Journal of Banking & Finance*, 34(8):1886–1898.

Bernanke, B. S. and Kuttner, K. N. (2005). What explains the stock market's reaction to federal reserve policy? *The Journal of finance*, 60(3):1221–1257.

Bouri, E., Azzi, G., and Dyhrberg, A. (2017). On the return–volatility relationship in the bitcoin market around the price crash of 2013. *Economics Letters*, 150:52–55.

Brøgger, S. (2022). Dynamic risk management and asset comovement. *Journal of Empirical Finance*, 67:60–77.

Chow, G. C. (1960). Tests of equality between sets of coefficients in two linear regressions. *Econometrica: Journal of the Econometric Society*, pages 591–605.

Christiano, L. J., Eichenbaum, M., and Evans, C. L. (2005). Nominal rigidities and the dynamic effects of a shock to monetary policy. *Journal of political Economy*, 113(1):1–45.

Egorov, M. (2019). Stableswap-efficient mechanism for stablecoin liquidity. *Retrieved Feb*, 24:2021.

Gudgeon, L., Werner, S., Perez, D., and Knottenbelt, W. J. (2020). Defi protocols for loanable funds: Interest rates, liquidity and market efficiency. In *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies*, pages 92–112.

Gürkaynak, R. S., Sack, B., and Swanson, E. (2005). Do actions speak louder than words? the response of asset prices to monetary policy actions and statements. *International Journal of Central Banking*, 1(1):55–93.

Krishnamurthy, A. and Vissing-Jorgensen, A. (2012). The aggregate demand for treasury debt. *Journal of Political Economy*, 120(2):233–267.

Markowitz, H. (1952). Modern portfolio theory. *Journal of Finance*, 7(11):77–91.

Nagel, S. (2016). The liquidity premium of near-money assets. *The Quarterly Journal of Economics*, 131(4):1927–1971.

Rahman, A., Shi, V., Ding, M., and Choi, E. (2022). Systematization of knowledge: Synthetic assets, derivatives, and on-chain portfolio management. *arXiv preprint arXiv:2209.09958*.

Schär, F. (2021a). Decentralized finance: On blockchain- and smart contract-based financial markets. *Federal Reserve Bank of St. Louis Review*, 103(2):153–174.

Schär, F. (2021b). Decentralized finance: On blockchain-and smart contract-based financial markets. *FRB of St. Louis Review*.

Sims, C. A., Stock, J. H., and Watson, M. W. (1990). Inference in linear time series models with some unit roots. *Econometrica*, 58(1):113–144.